

Fine-Tuning Language Models from Human Preferences

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Abstract

Reward learning enables the application of reinforcement learning (RL) to tasks where reward is defined by human judgment, building a model of reward by asking humans questions. Most work on reward learning has used simulated environments, but complex information about values is often expressed in natural language, and we believe reward learning for language is a key to making RL practical and safe for real-world tasks. In this paper, we build on advances in generative pretraining of language models to apply reward learning to four natural language tasks: continuing text with positive sentiment or physically descriptive language, and summarization tasks on the TL;DR and CNN/Daily Mail datasets. For stylistic continuation we achieve good results with only 5,000 comparisons evaluated by humans. For summarization, models trained with 60,000 comparisons copy whole sentences from the input but skip irrelevant preamble; this leads to reasonable ROUGE scores and very good performance according to our human labelers, but may be exploiting the fact that labelers rely on simple heuristics.

1. Introduction

We would like to apply reinforcement learning to complex tasks defined only by human judgment, where we can only tell whether a result is good or bad by asking humans. To do this, we can first use human labels to train a model of reward, and then optimize that model. While there is a long history of work learning such models from humans through interaction, this work has only recently been applied to modern deep learning, and even then has only been applied to relatively simple simulated environments (Christiano et al., 2017; Ibarz et al., 2018; Bahdanau et al., 2018). By contrast, real world settings in which humans need to specify com-

plex goals to AI agents are likely to both involve and require natural language, which is a rich medium for expressing value-laden concepts. Natural language is particularly important when an agent must communicate back to a human to help provide a more accurate supervisory signal (Irving et al., 2018; Christiano et al., 2018; Leike et al., 2018).

Natural language processing has seen substantial recent advances. One successful method has been to pretrain a large generative language model on a corpus of unsupervised data, then fine-tune the model for supervised NLP tasks (Dai and Le, 2015; Peters et al., 2018; Radford et al., 2018; Khandelwal et al., 2019). This method often substantially outperforms training on the supervised datasets from scratch, and a single pretrained language model often can be fine-tuned for state of the art performance on many different supervised datasets (Howard and Ruder, 2018). In some cases, fine-tuning is not required: Radford et al. (2019) find that generatively trained models show reasonable performance on NLP tasks with no additional training (zero-shot).

There is a long literature applying reinforcement learning to natural language tasks. Much of this work uses algorithmically defined reward functions such as BLEU for translation (Ranzato et al., 2015; Wu et al., 2016), ROUGE for summarization (Ranzato et al., 2015; Paulus et al., 2017; Wu and Hu, 2018; Gao et al., 2019b), music theory-based rewards (Jaques et al., 2017), or event detectors for story generation (Tambwekar et al., 2018). Nguyen et al. (2017) used RL on BLEU but applied several error models to approximate human behavior. Wu and Hu (2018) and Cho et al. (2019) learned models of coherence from existing text and used them as RL rewards for summarization and long-form generation, respectively. Gao et al. (2019a) built an interactive summarization tool by applying reward learning to one article at a time. Experiments using human evaluations as rewards include Kreutzer et al. (2018) which used off-policy reward learning for translation, and Jaques et al. (2019) which applied the modified Q-learning methods of Jaques et al. (2017) to implicit human preferences in dialog. Yi et al. (2019) learned rewards from humans to fine-tune dialog models, but smoothed the rewards to allow supervised learning. We refer to Luketina et al. (2019) for a survey of

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基于人类偏好对语言模型进行微调

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摘要

奖励学习使得强化学习（RL）能够被应用于那些奖励由人类判断所确定的任务，其实现方式是通过向人类提出问题来构建奖励模型。目前大多数关于奖励学习的研究都使用模拟环境，但实际上与价值相关的复杂信息往往以自然语言的形式存在，因此我们认为针对语言的奖励学习是让强化学习在现实世界任务中得以实用且安全应用的关键。在本文中，我们依托语言模型生成预训练领域的最新进展，将奖励学习应用于四类自然语言任务：生成带有积极情感或描述性语言的续写内容，以及针对TL;DR和CNN/Daily Mail数据集的摘要生成任务。在风格延续任务中，仅通过5000次由人类评估的对比即可获得良好效果。而在摘要生成任务中，经过60000次对比训练的模型虽然会直接复制输入中的完整句子并省略无关的前置内容，从而获得不错的ROUGE分数以及根据我们的人工标注员评估所得出的优异性能，但这可能是因为标注员在评分时依赖了简单的规则而已。

1. 引言

我们希望将强化学习应用于那些仅依靠人类判断来定义的复杂任务，在这类任务中，我们只能通过询问人类来判断结果的好坏。为此，我们可以先利用人类标注来训练一个用于评估奖励的模型，再对该模型进行优化。虽然通过人与模型的交互来学习此类模型已有悠久的历史，但这项技术直到近期才被应用到现代深度学习领域，而且最初也仅用于相对简单的模拟环境（Christiano等人，2017年；Ibarz等人，2018年；Bahdanau等人，2018年）。相比之下，现实中人类需要为人工智能代理指定复杂目标的情形，则很可能既涉及自然语言，又需要对自然语言有所要求，而自然语言正是表达蕴含价值的概念的理想载体。当人工智能代理必须向人类反馈信息以帮助提供更准确的监督信号时，自然语言的重要性更为凸显（

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自然语言对于表达蕴含价值的概念而言是一种极为丰富的媒介。尤其当人工智能代理需要向人类反馈信息，以便人类能够给出更精准的监督信号时，自然语言的作用就显得尤为重要（Irving等人，2018年；Christiano等人，2018年；Leike等人，2018年）。

近年来，自然语言处理领域取得了显著进展。其中一种有效的方法是先在无监督数据集上预训练一个大型生成型语言模型，然后再对该模型进行微调，以便用于有监督自然语言处理任务（Dai和Le，2015年；Peters等人，2018年；Radford等人，2018年；Khandelwal等人，2019年）。这种方法的表现往往远优于从零开始在有监督数据集上训练的模型，而且一个预训练语言模型通常只需稍加微调，就能在众多不同的有监督数据集上达到业界领先的性能水平（Howard和Ruder，2018年）。在某些情况下，则无需进行微调：Radford等人（2019年）发现，经过生成式训练的模型即便不进行额外训练，也能在自然语言处理任务中展现出不错的性能，即实现零样本学习。

已有大量文献探讨将强化学习应用于自然语言任务。这些研究大多采用通过算法定义的奖励函数，例如用于翻译任务的BLEU分数（Ranzato等人，2015年；Wu等人，2016年），用于摘要生成任务的ROUGE分数（Ranzato等人，2015年；Paulus等人，2017年；Wu和Hu，2018年；Gao等人，2019b年），基于音乐理论的奖励机制（Jaques等人，2017年），以及用于故事生成的事件检测器（Tambwekar等人，2018年）。Nguyen等人（2017年）在翻译任务中运用了强化学习，并结合多种误差模型来模拟人类行为。Wu和Hu（2018年）与Cho等人（2019年）分别从现有文本中学习连贯性模型，并将其作为摘要生成和长文本生成任务的强化学习奖励。Gao等人（2019a年）则通过一次处理一篇文章的方式，利用奖励学习构建出了交互式摘要工具。此外，也有研究以人类评估结果作为奖励，比如Kreutzer等人（2018年）采用离线策略强化学习进行翻译任务研究，而Jaques等人（2019年）则将Jaques等人（2017年）提出的改进后的Q学习方法应用于对话场景中的人类隐含偏好分析。Yi等人（2019年）通过从人类那里获取奖励来微调对话模型，但会对这些奖励进行平滑处理以便于开展监督学习。有关相关研究的综述，可参考Luketina等人的成果。

RL tasks involving language as a component, and for RL results using transfer learning from language. RL is not the only way to incorporate ongoing human feedback: [Hancock et al. \(2019\)](#) ask humans what a dialogue system should have said instead, then continue supervised training.

In this paper, we combine the pretraining advances in natural language processing with human preference learning. We fine-tune pretrained language models with reinforcement learning rather than supervised learning, using a reward model trained from human preferences on text continuations. Following [Jaques et al. \(2017; 2019\)](#), we use a KL constraint to prevent the fine-tuned model from drifting too far from the pretrained model. We apply our method to two types of tasks: continuing text in a way that matches a target style, either positive sentiment or vividly descriptive, and summarizing text from the CNN/Daily Mail or TL;DR datasets ([Hermann et al., 2015](#); [Völske et al., 2017](#)). Our motivation is NLP tasks where supervised data sets are unavailable or insufficient, and where programmatic reward functions are poor proxies for our true goals.

For stylistic continuation, 5,000 human comparisons (each choosing the best of 4 continuations) result in the fine-tuned model being preferred by humans 86% of the time vs. zero-shot and 77% vs. fine-tuning to a supervised sentiment network. For summarization, we use 60,000 human samples to train models that can roughly be described as “smart copiers”: they typically copy whole sentences from the input, but vary what they copy to skip irrelevant initial text. This copying behavior emerged naturally from the data collection and training process; we did not use any explicit architectural mechanism for copying as in [See et al. \(2017\)](#); [Gehrmann et al. \(2018\)](#). One explanation is that copying is an easy way to be accurate, given that we did not instruct labelers to penalize copying but do instruct them to penalize inaccuracy. It may also reflect the fact that some labelers check for copying as a fast heuristic to ensure a summary is accurate. Indeed, human labelers significantly prefer our models to supervised fine-tuning baselines and even to human-written reference summaries, but not to a lead-3 baseline which copies the first three sentences.

For summarization, we continue to collect additional data and retrain our reward model as the policy improves (*online* data collection). We also test *offline* data collection where we train the reward model using data from the original language model only; offline data collection significantly reduces the complexity of the training process. For the TL;DR dataset, human labelers preferred the policy trained with online data collection 71% of the time, and in qualitative evaluations the offline model often provides inaccurate summaries. In contrast, for stylistic continuation we found that offline data collection worked similarly well. This may be related to the style tasks requiring very little data; [Radford](#)

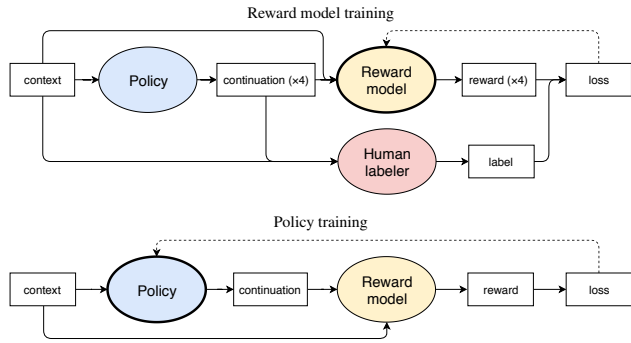


Figure 1: Our training processes for reward model and policy. In the online case, the processes are interleaved.

[et al. \(2017\)](#) show that generatively trained models can learn to classify sentiment from very few labeled examples.

In concurrent work, [Böhm et al. \(2019\)](#) also use human evaluations to learn a reward function for summarization, and optimize that reward function with RL. Their work provides a more detailed investigation of the learned policy and reward function on the CNN/Daily Mail dataset, while we are interested in exploring learning from human feedback more generally and at larger computational scale. So we consider several additional tasks, explore the effects of on-policy reward model training and more data, and fine-tune large language models for both reward modeling and RL.

2. Methods

We begin with a vocabulary Σ and a language model ρ which defines a probability distribution over sequences of tokens Σ^n via

$$\rho(x_0 \cdots x_{n-1}) = \prod_{0 \leq k < n} \rho(x_k | x_0 \cdots x_{k-1})$$

We will apply this model to a task with input space $X = \Sigma^{\leq m}$, data distribution $\mathcal{D}$ over X , and output space $Y = \Sigma^n$. For example, $x \in X$ could be an article of up to 1000 words and $y \in Y$ could be a 100-word summary. ρ defines a probabilistic policy for this task via $\rho(y|x) = \rho(xy)/\rho(x)$: fixing the beginning of the sample to x and generating subsequent tokens using ρ .

We initialize a policy $\pi = \rho$, and then fine-tune π to perform the task well using RL. If the task was defined by a reward function $r : X \times Y \rightarrow \mathbb{R}$, then we could use RL to directly optimize the expected reward:

$$\mathbb{E}_\pi [r] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x)} [r(x, y)]$$

However, we want to perform tasks defined by human judgments, where we can only learn about the reward by asking humans. To do this, we will first use human labels to train a reward model, and then optimize that reward model.

那些以语言作为组成部分的强化学习任务，以及那些借助语言领域的迁移学习所得出的强化学习研究成果。不过，强化学习并非整合人类实时反馈的唯一途径：[Hancock等人\(2019年\)](#)会先询问人类在面对某种对话系统时应给出怎样的回应，随后再继续进行监督学习。

在本文中，我们将自然语言处理领域的预训练技术进展与人类偏好学习相结合。我们并未采用监督学习，而是借助基于人类偏好训练得到的奖励模型，通过强化学习对预训练语言模型进行微调。与[Jaques等人\(2017年; 2019年\)](#)的研究思路不同，我们引入了KL约束，以防止微调后的模型与预训练模型出现过大大偏差。该方法被应用于两类任务：一是以符合特定风格的方式续写文本，这类风格包括积极情感表达或生动描述性风格；二是对来自CNN/《每日邮报》或TL;DR数据集的文本进行摘要生成([Hermann等人, 2015年](#); [Völske等人, 2017年](#))。我们开展这项研究的动机，是针对那些缺乏足够监督学习数据集，或是程序化奖励函数无法有效反映真实目标的自然语言处理任务。

在风格化文本续写任务中，通过5000次人类对比实验（每次由人类从4种续写结果中选出最佳选项），我们发现微调后的模型有86%的概率更受人类青睐，而零样本方法的这一比例仅为0%，相比针对监督学习情感网络的微调方法，这一比例则为77%。在摘要生成任务中，我们使用了60000份人类样本来训练模型，这类模型大致可被称为“智能复制器”：它们通常会复制输入文本中的整句内容，但会通过调整复制内容来跳过无关的开头文字。这种复制行为是在数据收集和训练过程中自然产生的，我们并未像[See等人\(2017年\)](#)；[Gehrmann等人\(2018年\)](#)的研究那样，为复制行为设计任何特定的架构机制。其中一个可能的解释是，由于我们没有要求标注员对复制行为给予惩罚，只要求他们对不准确之处进行惩罚，因此复制成为了一种实现高准确率简便方式。这也可能反映出部分标注员会通过检查是否存在复制行为作为一种快速判断手段，以确保摘要的准确性。事实上，人工标注员显著更倾向于选择我们的模型，而非基于监督学习的微调基线模型，甚至优于人工撰写的参考摘要，但却不如会复制前三句话的Lead-3基线模型。

在开展摘要生成工作时，我们会持续收集更多数据，并随着策略的不断优化而重新训练我们的奖励模型。

在线模式下的数据收集。此外，我们还会测试离线模式下的数据收集方式——在这种模式下，我们仅使用原始语言模型产生的数据来训练奖励模型；离线数据收集能够显著降低训练过程的复杂性。在TL;DR数据集上，有71%的情况下人工标注员更倾向于选择通过在线数据收集训练得到的策略，而且在定性评估中，离线模型往往无法给出准确的总结。相比之下，在进行风格化文本续写任务时，我们发现离线数据收集的效果也相当不错。这可能是因为这类风格相关任务对数据量要求极低所致；[Radford](#)

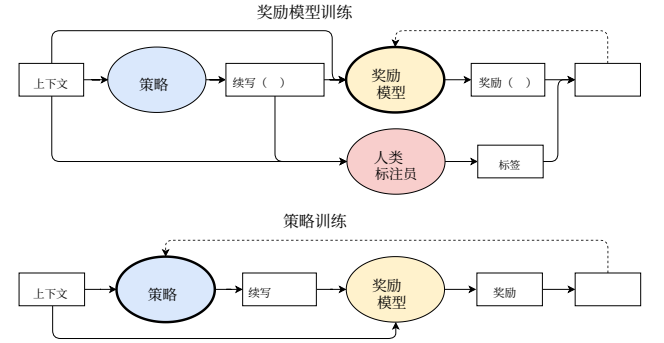


图1：我们用于奖励模型与策略训练的流程。在在线模式下，这两种训练流程会交替进行。

[等人\(2017年\)](#)的研究表明，通过生成式方法训练得到的模型即便只有极少量的带标签样本，也能学会对情感倾向进行分类。

在相关的其他研究中，[Böhm等人（2019年）](#)也借助人类评估结果来构建用于摘要生成的奖励函数，并通过强化学习对该奖励函数进行优化。他们的研究更深入地分析了在CNN/每日邮报数据集上训练得到的策略与奖励函数，而我们的目标则是更广泛地探索如何从人类反馈中学习，且要在更大的计算规模下开展相关研究。因此，我们考虑了若干额外的任务，研究了在策略训练过程中融入奖励模型以及增加数据量所带来的影响，同时还对大型语言模型进行了微调，以便同时用于奖励建模和强化学习领域。

2. 方法

我们首先从一个词汇表 Σ 以及一个语言模型 ρ 开始，该语言模型通过某种方式定义了词元序列 Σ^n 上的概率分布。

$$\rho(x_0 \cdots x_{n-1}) = \prod_{0 \leq k < n} \rho(x_k | x_0 \cdots x_{k-1})$$

我们将把这个模型应用到这样一类任务中：其输入空间为 $X = \Sigma^{\leq m}$ ，数据分布在 $\mathcal{D}$ 之上且服从 X 的分布，输出空间则为 $Y = \Sigma^n$ 。例如， $x \in X$ 可以是一篇长度不超过1000字的文章，而 $y \in Y$ 则可以是长度为100字的摘要。 ρ 通过 $\rho(y|x) = \rho(xy)/\rho(x)$ 的方式为这类任务定义了概率策略：即固定样本的开头部分为 x ，然后利用 ρ 来生成后续的词元。

我们首先初始化一个策略 $\pi = \rho$ ，然后通过强化学习对其 π 进行微调，从而更高效地完成该任务。如果该任务是由奖励函数 $r : X \times Y \rightarrow \mathbb{R}$ 来定义的，那么我们就可以利用强化学习直接优化预期奖励值。

$$\mathbb{E}_\pi [r] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x)} [r(x, y)]$$

然而，我们希望执行由人类判断所定义的任务，在这类任务中只能通过询问人类来了解对应的奖励信息。为此，我们首先会利用人类标注来训练一个奖励模型，随后再对该奖励模型进行优化。

Following [Christiano et al. \(2017\)](#), we ask human labelers to pick which of several values of y_i is the best response to a given input x .¹ We ask humans to choose between four options (y_0, y_1, y_2, y_3) ; considering more options allows a human to amortize the cost of reading and understanding the prompt x . Let $b \in \{0, 1, 2, 3\}$ be the option they select. Having collected a dataset S of $(x, y_0, y_1, y_2, y_3, b)$ tuples, we fit a reward model $r : X \times Y \rightarrow \mathbb{R}$ using the loss

$$\text{loss}(r) = \mathbb{E}_{(x, \{y_i\}_i, b) \sim S} \left[\log \frac{e^{r(x, y_b)}}{\sum_i e^{r(x, y_i)}} \right] \quad (1)$$

Since the reward model needs to understand language, we initialize it as a random linear function of the final embedding output of the language model policy ρ following [Radford et al. \(2018\)](#) (see [section 4.2](#) for why we initialize from ρ rather than π). To keep the scale of the reward model consistent across training, we normalize it so that it has mean 0 and variance 1 for $x \sim \mathcal{D}, y \sim \rho(\cdot|x)$.

Now we fine-tune π to optimize the reward model r . To keep π from moving too far from ρ , we add a penalty with expectation $\beta \text{KL}(\pi, \rho)$ (see [table 10](#) for what happens without this). That is, we perform RL on the modified reward

$$R(x, y) = r(x, y) - \beta \log \frac{\pi(y|x)}{\rho(y|x)}. \quad (2)$$

We either choose a constant β or vary it dynamically to achieve a particular value of $\text{KL}(\pi, \rho)$; see [section 2.2](#). This term has several purposes: it plays the role of an entropy bonus, it prevents the policy from moving too far from the range where r is valid, and in the case of our style continuation tasks it also is an important part of the task definition: we ask humans to evaluate style, but rely on the KL term to encourage coherence and topicality.

Our overall training process is:

1. Gather samples (x, y_0, y_1, y_2, y_3) via $x \sim \mathcal{D}, y_i \sim \rho(\cdot|x)$. Ask humans to pick the best y_i from each.
2. Initialize r to ρ , using random initialization for the final linear layer of r . Train r on the human samples using loss [\(1\)](#).
3. Train π via Proximal Policy Optimization (PPO, [Schulman et al. \(2017\)](#)) with reward R from [\(2\)](#) on $x \sim \mathcal{D}$.
4. In the online data collection case, continue to collect additional samples, and periodically retrain the reward model r . This is described in [section 2.3](#).

¹In early experiments we found that it was hard for humans to provide consistent fine-grained quantitative distinctions when asked for an absolute number, and experiments on synthetic tasks confirmed that comparisons were almost as useful.

2.1. Pretraining details

We use a 774M parameter version of the GPT-2 language model in [Radford et al. \(2019\)](#) trained on their WebText dataset and their 50,257 token invertible byte pair encoding to preserve capitalization and punctuation ([Sennrich et al., 2015](#)). The model is a Transformer with 36 layers, 20 heads, and embedding size 1280 ([Vaswani et al., 2017](#)).

For stylistic continuation tasks we perform supervised fine-tuning of the language model to the BookCorpus dataset of [Zhu et al. \(2015\)](#) prior to RL fine-tuning; we train from scratch on WebText, supervised fine-tune on BookCorpus, then RL fine-tune to our final task. To improve sample quality, we use a temperature of $T < 1$ for all experiments; we modify the initial language model by dividing logits by T , so that future sampling and RL with $T = 1$ corresponds to a lower temperature for the unmodified pretrained model.

2.2. Fine-tuning details

Starting with the pretrained language model, the reward model is trained using the Adam optimizer ([Kingma and Ba, 2014](#)) with loss [\(1\)](#). The batch size is 8 for style tasks and 32 for summarization, and the learning rate is 1.77×10^{-5} for both. We use a single epoch to avoid overfitting to the small amount of human data, and turn off dropout.

For training the policy π , we use the PPO2 version of Proximal Policy Optimization from [Dhariwal et al. \(2017\)](#). We use 2M episodes (x, y) pairs, $\gamma = 1$, four PPO epochs per batch with one minibatch each, and default values for the other parameters. We use batch size 1024 for style tasks and 512 for summarization. We do not use dropout for policy training. The learning rate was 1.41×10^{-5} for style tasks and 7.07×10^{-6} for summarization.

Models trained with different seeds and the same KL penalty β sometimes end up with quite different values of $\text{KL}(\pi, \rho)$, making them hard to compare. To fix this, for some experiments we dynamically vary β to target a particular value of $\text{KL}(\pi, \rho)$ using the log-space proportional controller

$$e_t = \text{clip} \left(\frac{\text{KL}(\pi_t, \rho) - \text{KL}_{\text{target}}}{\text{KL}_{\text{target}}}, -0.2, 0.2 \right) \\ \beta_{t+1} = \beta_t (1 + K_\beta e_t)$$

We used $K_\beta = 0.1$.

For supervised fine-tuning baselines, we fine-tune for 1 epoch on the CNN/Daily Mail and TL;DR training sets (for TL;DR we removed 30K examples to serve as a validation set). We decayed the learning rate to 0 with a cosine schedule; for the initial value, we swept over 8 log-linearly spaced options between 10^{-4} and 3×10^{-4} . We also experimented with different dropout rates, and found a rate of 0.1 to work best. We then chose the model with the best validation loss.

遵循Christiano等人（2017年）的研究方法，我们会让人工标注员从多个 y_i 取值中选出针对给定输入 x ¹ 的最佳响应。我们会要求他们在四个选项 (y_0, y_1, y_2, y_3) 中做出选择；提供更多选项有助于人工标注员分摊阅读和理解提示 x 所需的时间成本。设 $b \in \{0, 1, 2, 3\}$ 为他们选中的选项。在收集到包含 $(x, y_0, y_1, y_2, y_3, b)$ 元组的数据库 S 后，我们将利用损失函数来拟合奖励模型 $r : X \times Y \rightarrow \mathbb{R}$ 。

$$\text{loss}(r) = \mathbb{E}_{(x, \{y_i\}_i, b) \sim S} \left[\log \frac{e^{r(x, y_b)}}{\sum_i e^{r(x, y_i)}} \right] \quad (1)$$

由于奖励模型需要理解语言，我们将其初始化为语言模型策略的最终嵌入输出的随机线性函数 ρ 依据[Radford等人\(2018年\)](#)（具体原因请参见[4.2节](#)，即为何我们要从 ρ 而非 π 进行初始化）。为确保训练过程中奖励模型的尺度保持一致，我们会对其进行归一化处理，使其在 $x \sim \mathcal{D}, y \sim \rho(\cdot|x)$ 时的均值為0且方差為1。

现在我们对 π 进行微调，以优化 r 奖励模型 π ，避免其偏离 ρ 过远。为此，我们引入了一个带期望值的惩罚项 $\beta \text{KL}(\pi, \rho)$ （具体内容可参见[表10](#)，了解不使用该惩罚项时的情况）。也就是说，我们是在经过修改的奖励函数上执行强化学习的

$$R(x, y) = r(x, y) - \beta \log \frac{\pi(y|x)}{\rho(y|x)}. \quad (2)$$

我们可以选择一个常数 β ，或者动态调整该常数的值，以此使得 $\text{KL}(\pi, \rho)$ 的值达到特定数值；详情请参见[2.2节](#)。这一项具有多种作用：它既可作为熵奖励，防止策略偏离 r 有效范围过远，同时在我们的风格续写任务中，它也是任务定义的重要组成部分——我们虽然要求人类对风格进行评价，但会借助KL项来提升文本的连贯性与与主题的相关性。

我们的整体训练流程如下：

1. 通过 $x \sim \mathcal{D}, ()$ 等方式收集样本（、 y_0, y_1, y_2, y_3 ）。随后请人类从每组样本中挑选出最优的 y_i 选项。
2. 初始化 r 为 ρ ，其中最终线性层采用随机初始化方式，即。接着使用损失函数<style id='15'(1)在人类标注的样本上对 r 进行训练。
3. 采用近端策略优化算法（PPO, [Schulman等人，（2017年）](#)根据 [（2）中的奖励机制](#)在 $x \sim \mathcal{D}$ 上进行训练。
4. 在在线数据收集场景中，需持续采集更多样本，并定期重新训练奖励模型 r 。相关内容可见[第2.3节](#)。

¹在早期的实验中我们发现，当要求人类给出绝对数值时，他们很难提供一致且精细的定量对比；而在合成任务上的实验也表明，此类比较几乎不具备参考价值。

2.1. 预训练细节

我们在此采用了由Radford等人[在2019年](#)基于其WebText数据集以及采用50,257个令牌的可逆字节对编码方式来保留大小写与标点符号的技术（该技术由Sennrich等人于2015年提出）所训练的、拥有774百万参数的GPT-2语言模型版本。[该模型属于Transformer架构，共包含36层、20个注意力头，嵌入维度为1280（这一设计由Vaswani等人于2017年提出）。](#)

在开展风格化文本续写任务时，我们首先会对语言模型进行监督学习微调，使其适配[Zhu等人（2015）](#)构建的BookCorpus语料库，之后再开展强化学习微调；具体流程为：先在WebText数据上从零开始训练模型，接着在BookCorpus语料库上进行监督学习微调，最后通过强化学习微调来完成最终任务。为提升样本质量，所有实验都会设置 $T < 1$ 特定的温度参数；同时我们会通过将逻辑值除以 T 的方式对初始语言模型进行修改，这样后续的采样过程以及基于 $T = 1$ 该模型的强化学习训练，其效果就相当于未做修改的预训练模型使用了更低的温度参数。

2.2. 微调细节

以预训练语言模型为基础，奖励模型会采用Adam优化器（[Kingma和Ba，2014](#)）进行训练，损失函数为[\(1\)](#)。对于风格化任务，批量大小为8，而摘要生成任务的批量大小则为32，两者的学习率都设置为 1.77×10^{-5} 。为避免因人类标注数据量较少而导致过拟合，我们仅训练一个周期，并且关闭dropout机制。

在训练策略时 π ，我们采用了Dhariwal等人[在（2017年）](#)提出的Proximal Policy Optimization的PPO2版本，每批处理包含200万剧集对，每批进行四轮PPO训练，每轮使用一个小型数据子集，其余参数则采用默认值。对于风格相关任务，我们设置批量大小为1024；而对于摘要生成任务，则设置为512。在策略训练过程中我们不使用dropout机制。风格相关任务的学习率设为 1.41×10^{-5} ，而摘要生成任务的学习率则设为 7.07×10^{-6} 。

使用不同种子但施加相同KL惩罚的模型 β ，其KL值 π 有时会呈现出相当大的差异 ρ ，这使得这些模型难以相互比较。为了解决这个问题，在部分实验中我们会通过对 β 进行动态调整，并利用对数空间比例控制器，将KL值控制在与目标值 π 相符的范围之内 ρ 。

$$e_t = \text{clip} \left(\frac{\text{KL}(\pi_t, \rho) - \text{KL}_{\text{target}}}{\text{KL}_{\text{target}}}, -0.2, 0.2 \right) \\ \beta_{t+1} = \beta_t (1 + K_\beta e_t)$$

我们使用了 $K_\beta = 0.1$ 。

对于基于监督学习的微调基准方案，我们在CNN/每日邮报以及TL;DR训练集上进行了1个时代的微调（针对TL;DR训练集，我们剔除了3万条样本作为验证集）。我们采用余弦调度策略将学习率逐步降至0，初始值则在 10^{-4} 与 3×10^{-4} 之间选取了8个对数线性间隔的选项进行测试。此外，我们还尝试了不同的丢弃率，最终发现0.1的丢弃率效果最佳。最后，我们选择了在验证集上损失最小的模型。

2.3. Online data collection

If the trained policy π is very different from the zero-shot policy ρ , the reward model will suffer a large distributional shift from training on samples from ρ to evaluation on samples from π . To prevent this, we can collect human data throughout RL fine-tuning, continuously gathering new data by sampling from π and retraining the reward model. As [section 3](#) shows, online data collection was important for summarization but not for the simpler style tasks.

In the online case, we will choose a function $l(n)$ describing how many labels we want before beginning the n^{th} PPO episode. Let $N_\pi = 2 \times 10^6$ be the total number of PPO episodes, $N_r^0 = l(0)$ be an initial number of human labels, and N_r be the total number of human labels. We take

$$l(n) = N_r^0 + (N_r - N_r^0) (1 - (1 - n/N_\pi)^2)$$

We pause before the n^{th} PPO episode if we have fewer than $l(n)$ labels. We send another batch of requests to the labelers if the total requests so far is less than $l(n) + 1000$, to ensure they have at least 1000 outstanding queries at any time. We train the reward model before the first PPO episode, and then retrain it 19 more times at evenly spaced values of $l(n)$. Each time we retrain we reinitialize r to a random linear layer on top of ρ and do a single epoch through the labels collected so far. The offline case is the limit $N_r = N_r^0$.

To estimate overall progress, we gather validation samples consisting of $x \sim \mathcal{D}; y_0, y_1 \sim \rho(\cdot|x); y_2, y_3 \sim \pi(\cdot|x)$ at a constant rate; human labels on these give how often π beats ρ . Since validation samples are only used to evaluate the current π , we can add them to the training set for r . In order to estimate inter-labeler agreement, 5% of queries are answered 5 times by different labelers. Label counts in [section 3](#) include validation samples and repeated labels.

2.4. Human labeling

We use [Scale AI](#) to collect labels. The Scale API accepts requests of the form (x, y_0, y_1, y_2, y_3) and returns selections $b \in \{0, 1, 2, 3\}$. We describe the task to Scale through a combination of instructions ([appendix A](#)) and a dataset of about 100 example comparisons from the authors.

Unlike many tasks in ML, our queries do not have unambiguous ground truth, especially for pairs of similar outputs (which play a large role in our training process, since we train r on pairs of labels sampled from a single policy π). This means that there is significant disagreement even between labelers who have a similar understanding of the task and are trying to rate consistently. On 4-way comparisons for sentiment and TL;DR summarization, authors of this paper agree about 60% of the time (vs. 25% for random guessing). This low rate of agreement complicates the quality control process for Scale; the authors agree with Scale

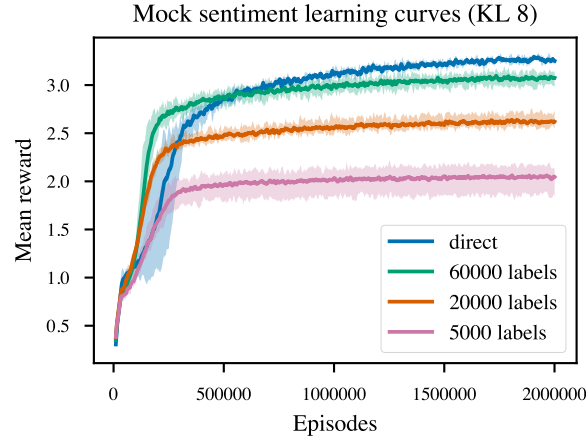


Figure 2: Learning curves for a 124M-parameter model with mock sentiment reward, targeting a KL of 8 nats. Lines and shaded areas show mean and range for 5 seeds. Early on the reward model sometimes speeds up training, a phenomenon also observed by [Christiano et al. \(2017\)](#).

labelers 38% of the time on sentiment and 46% of the time on TL;DR summarization. We give further details of the human data collection and quality evaluation in [appendix B](#).

For final evaluation of two models A and B , we generate either 2-way comparisons between pairs $(a \sim A, b \sim B)$ or 4-way comparisons with quadruples $(a_0, a_1 \sim A, b_0, b_1 \sim B)$, randomize the order in which samples are presented, and present these comparisons to Scale. Evaluating the quality of a model trained by Scale using the same set of humans from Scale is perilous: it demonstrates that r and π have succeeded in fitting to the human reward, but does not show that those human evaluations capture what we really care about, and our models are incentivized to exploit idiosyncracies of the labeling process. We include samples from our models so that readers can judge for themselves.

3. Experiments

In [section 3.1.1](#), we test our approach to RL fine-tuning of language models by using a mock labeler (a sentiment model trained on a review classification problem) as a stand-in for human labels. We show that RL fine-tuning is effective at optimizing this complex but somewhat artificial reward. In [section 3.1.2](#), we show that we can optimize language models from human preferences on stylistic continuation tasks (sentiment and physical descriptiveness) with very little data, and that in the sentiment case the results are preferred to optimizing the review sentiment model. In [section 3.2](#) we apply RL fine-tuning to summarization on the CNN/Daily Mail and TL;DR datasets, show that the resulting models are essentially “smart copiers”, and discuss these results in the context of other summarization work.

2.3. 在线数据收集

如果经过训练得到的策略 π 与零样本策略存在极大差异 ρ , 那么奖励模型在从 ρ 训练时的样本过渡到 π 评估时的样本时, 就会出现严重的分布偏移。为避免这种情况, 我们可以在强化学习微调过程中持续收集人类产生的数据, 通过从 π 中采样不断获取新数据, 进而重新训练奖励模型。正如[第3节](#)所示, 在摘要生成任务中在线数据收集非常重要, 但对于较为简单的文本风格优化任务则并非必需。

在在线模式下, 我们会选定一个函数 $l(n)$, 用来确定在开始某次 n^{th} PPO训练轮次之前需要准备多少个标签。设 $N_\pi = 2 \times 10^6$ 为PPO训练轮次的总数, $N_r^0 = l(0)$ 为初始的人类标注数量, N_r 则为所有人类标注的总量, 我们依据这些数值来制定相应策略。

$$l(n) = N_r^0 + (N_r - N_r^0) (1 - (1 - n/N_\pi)^2)$$

如果现有的标签数量少于 $l(n)$, 我们就会在开始该次 n^{th} PPO训练轮次之前暂停操作。此外, 若迄今为止发送的请求总数低于 $l(n) + 1000$, 我们还会向标注员发送更多批次的请求, 以确保他们任何时候都有至少1000个待处理的查询。我们会在第一次PPO训练轮次开始前先训练奖励模型, 之后在 $l(n)$ 设定的等间隔时间点上对其重新训练19次。每次重新训练时, 我们都会将 r 重新初始化为 ρ 之上的一个随机线性层, 然后使用目前收集到的所有标签完成一个训练周期。而离线模式则是上述方法的极限情况 $N_r = N_r^0$ 。

为评估整体进展, 我们会以恒定速率收集由 $x \sim \mathcal{D}; y_0, y_1 \sim \rho(\cdot|x) y_2, y_3 \sim \pi(\cdot|x)$ 组成的验证样本; 同时也会获取针对这些样本的人类标注, 以此判断 π “出现次数” ρ 的频率。由于验证样本仅用于评估当前的 π 模型性能, 因此我们可以将其加入训练集, 以便用于 r 后续优化。为衡量不同标注员之间的标注一致性, 我们会让5%的查询由不同的标注员各标注5次。[第3节](#)中列出的标签出现次数统计包含了验证样本以及重复标注的记录。

2.4. 人类标注

我们使用[Scale AI](#) 来收集各类标签。Scale API能够接收如下格式的请求 (x, y_0, y_1, y_2, y_3) , 并返回相应的标签选择结果 $b \in \{0, 1, 2, 3\}$ 。我们会通过一系列指令 (参见[附录A](#)) 以及研究者们提供的约100组示例对比数据, 向Scale明确说明需要完成的任务。

与机器学习领域的许多任务不同, 我们的查询并不具备明确的真实值, 尤其是对于那些输出相似的配对数据而言 (这类数据在我们的训练过程中起着重要作用, 因为我们是通过从同一策略中抽取的标签对来 r 进行训练的 π)。这意味着, 即便那些对任务有相似理解且试图给出一致评分的标注员之间, 也存在着相当大的分歧。在针对情感倾向和简而言之摘要生成的四向对比实验中, 本文的研究者们仅有60%的时间能达成一致意见 (而随机猜测的准确率仅为25%)。如此低的一致性比例给Scale公司的质量控制工作带来了很大困难; 不过研究者们还是认同Scale公司的做法。

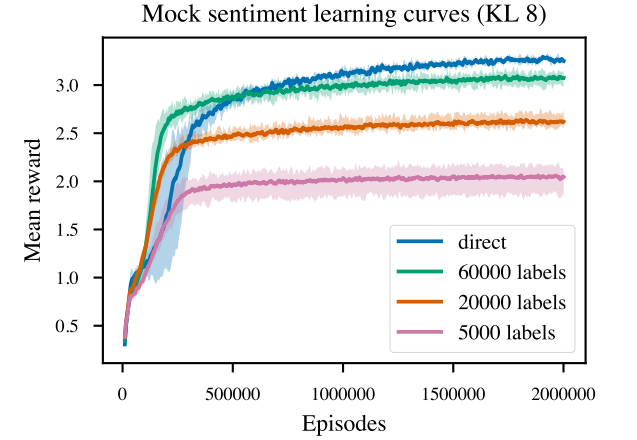


图2: 采用模拟情感奖励的1.24亿参数模型的学习曲线, 目标 KL散度为8纳特。曲线及阴影区域分别显示了5组随机种子对应的平均值与数值范围。在训练初期, 奖励模型有时会加速训练过程, 这一现象也已被[Christiano等人 \(2017年\)](#) 所观察到。

在情感倾向评估方面标注员的一致性率为38%, 而在简而言之摘要生成方面的这一比例则为46%。关于人类数据收集及质量评估的更多细节, 可见于[附录B](#)。

在对 A 和 B 这两种模型进行最终评估时, 我们会生成成对的双向对比 $(a \sim A, b \sim B)$, 或是包含四元组的四向对比 $(a_0, a_1 \sim A, b_0, b_1 \sim B)$, 同时随机安排样本的呈现顺序, 再将这些对比结果提交给Scale公司。不过, 使用Scale公司相同团队的人员来评估由该平台训练出的模型质量存在很大风险: 这类评估只能说明 r 和 π 已成功适配人类设定的奖励标准, 却无法证明这些人类评估结果真正反映了我们关注的核心指标, 同时也可能导致我们的模型刻意利用标注过程中的特殊规律来获取优势。为此, 我们在文中加入了自家模型的样本, 以便读者能够自行判断。

3. 实验

在[3.1.1节](#)中, 我们通过使用模拟标注器 (一种基于评论分类任务训练的情感模型) 来替代人类标注, 以此测试我们针对语言模型进行强化学习微调的方法。实验结果表明, 强化学习微调能够有效优化这种复杂但略带人为设定的奖励机制。在[3.1.2节](#)中, 我们证明了即便只有极少的数据, 也能利用人类偏好在风格化文本续写任务 (包括情感倾向与描述性表现方面) 上对语言模型进行优化, 并且在处理情感倾向任务时, 所得到的模型效果甚至优于专门优化评论情感模型的结果。在[3.2节](#)中, 我们将强化学习微调技术应用于CNN/每日邮报及TL;DR数据集上的摘要生成任务, 发现由此产生的模型本质上属于“智能复制器”类型, 同时还将这些结果与其他摘要生成相关研究进行了对比分析。

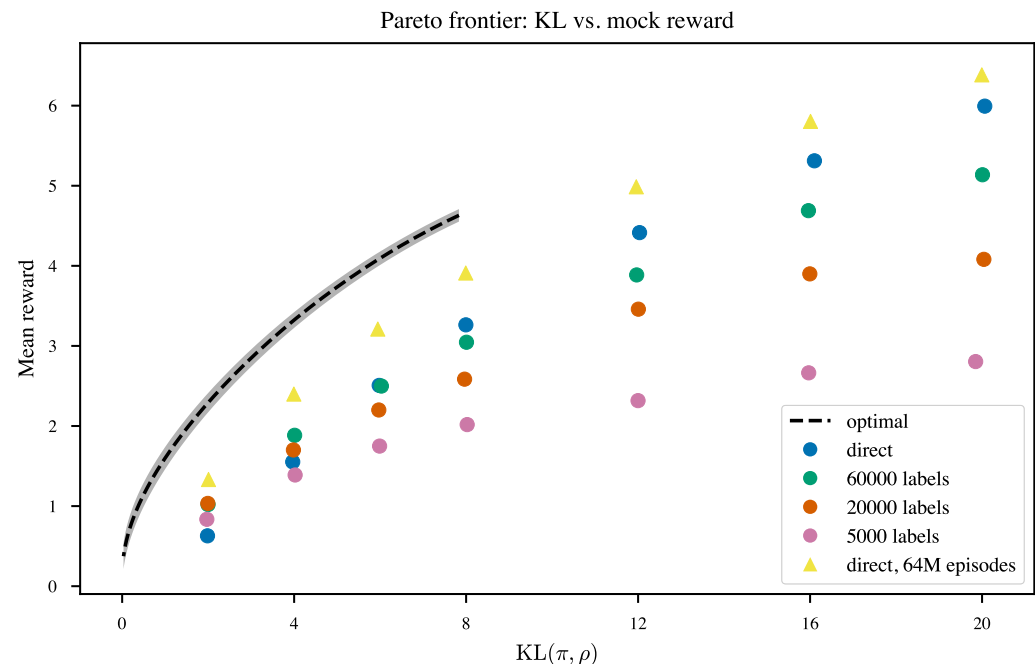


Figure 3: Allowing the policy π to move further from the initial policy ρ as measured by $KL(\pi, \rho)$ achieves higher reward at the cost of less natural samples. Here we show the optimal KL vs. reward for 124M-parameter mock sentiment (as estimated by sampling), together with results using PPO. Runs used 2M episodes, except for the top series.

We [release code](#)² for reward modeling and fine-tuning in the offline data case. Our public version of the code only works with a smaller 124M parameter model with 12 layers, 12 heads, and embedding size 768. We include fine-tuned versions of this smaller model, as well as some of the human labels we collected for our main experiments (note that these labels were collected from runs using the larger model).

3.1. Stylistic continuation tasks

We first apply our method to stylistic text continuation tasks, where the policy is presented with an excerpt from the BookCorpus dataset (Zhu et al., 2015) and generates a continuation of the text. The reward function evaluates the style of the concatenated text, either automatically or based on human judgments. We sample excerpts with lengths of 32 to 64 tokens, and the policy generates 24 additional tokens. We set the temperature of the pretrained model to $T = 0.7$ as described in [section 2.1](#).

3.1.1. MOCK SENTIMENT TASK

To study our method in a controlled setting, we first apply it to optimize a known reward function r_s designed to reflect some of the complexity of human judgments. We construct

r_s by training a classifier³ on a binarized, balanced subsample of the Amazon review dataset of McAuley et al. (2015). The classifier predicts whether a review is positive or negative, and we define $r_s(x, y)$ as the classifier’s log odds that a review is positive (the input to the final sigmoid layer).

Optimizing r_s without constraints would lead the policy to produce incoherent continuations, but as described in [section 2.2](#) we include a KL constraint that forces it to stay close to a language model ρ trained on BookCorpus.

The goal of our method is to optimize a reward function using only a small number of queries to a human. In this mock sentiment experiment, we simulate human judgments by assuming that the “human” always selects the continuation with the higher reward according to r_s , and ask how many queries we need to optimize r_s .

[Figure 2](#) shows how r_s evolves during training, using either direct RL access to r_s or a limited number of queries to train a reward model. 20k to 60k queries allow us to optimize r_s nearly as well as using RL to directly optimize r_s .

Because we know the reward function, we can also analytically compute the optimal policy and compare it to our learned policies. With a constraint on the KL divergence

³The model is a Transformer with 6 layers, 8 attention heads, and embedding size 512.

²Code at <https://github.com/openai/lm-human-preferences>.

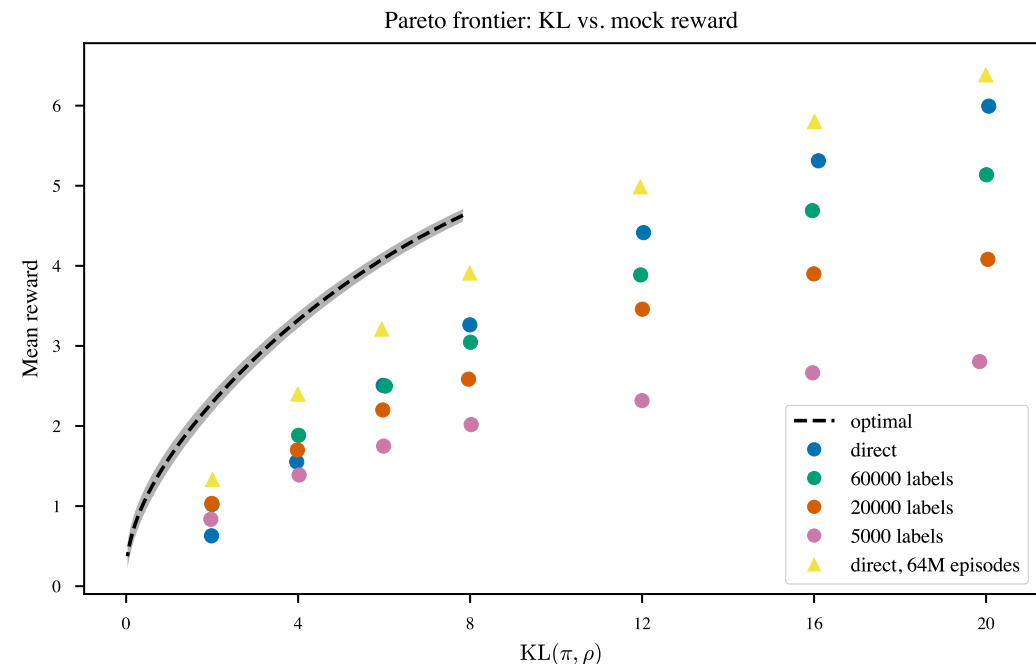


图3显示：通过KL散度(π, ρ)来衡量，让策略 π 偏离初始策略 ρ 的程度越大，虽然生成的样本自然度会降低，但获得的奖励却越高。此处我们展示了124百万参数的模拟情感数据（通过抽样估算）对应的最佳KL散度与奖励关系，同时也给出了使用PPO算法得到的结果。除最顶层的几组数据外，所有实验均使用了200万剧集的数据量。

我们[发布了](#)²用于离线数据场景下进行奖励建模与微调的代码。我们公开的代码版本仅适用于参数数量为1.24亿、具有12层结构、12个输出头且嵌入维度为768的较小模型。我们还提供了该较小模型的微调版本，以及为主实验收集的部分人类标注数据（需注意，这些标注数据是从使用较大模型进行的实验中获取的）。

3.1. 风格化文本续写任务

首先，我们将该方法应用于风格化文本续写任务中——在这种任务中，策略会接收到来自BookCorpus数据集的片段（Zhu等人，2015年），并据此生成文本的续写内容。奖励函数会自动或依据人类判断来评估拼接后文本的风格表现。我们选取长度在32到64个标记之间的片段作为输入，然后让策略再生成24个标记的续写内容。预训练模型的温度参数则设定为 $T = 0.7$ ，具体设置方式如[2.1节](#)所述。

3.1.1. 模拟情感倾向任务

为在受控环境中测试我们的方法，我们首先将其应用于优化一个已知的奖励函数 r_s ，该函数旨在体现人类判断中存在的一些复杂性。随后我们构建了

²相关代码可见于<https://github.com/openai/lm-human-preferences>。

r_s 通过训练分类器³在McAuley等人（2015）所提出的亚马逊评论数据集的二值化且均衡的子样本上。 $\phi(x, y)$ 被定义为该分类器预测某条评论为正面的对数几率（即传递给最终Sigmoid层的输入值）。

若在优化过程中不设置任何约束，策略就会生成逻辑混乱的续写结果；但正如[第2.2节](#)所述，我们会加入KL约束，迫使该策略的输出始终接近在BookCorpus语料库上训练得到的语言模型。

我们的方法旨在仅通过少量向人类咨询的次数，即可优化奖励函数。在这项模拟情感倾向的实验中，我们假设“人类评估者”会始终根据 r_s 选择奖励更高的续写结果，进而以此来探究完成优化需要多少次咨询 r_s 。

[图2](#)展示了在训练过程中的变化情况，这些变化是通过直接应用强化学习算法或通过有限次数的咨询来训练奖励模型而产生的。2万到6万次咨询的效率，几乎可以与直接使用强化学习进行优化相媲美。

由于我们已知奖励函数，因此也可以通过分析方法计算出最优策略，并将其与我们学习到的策略进行对比。在施加KL散度约束的前提下，

³该模型采用Transformer架构，共包含6层结构，配备8个注意力头数，嵌入维度为512。

Fine-Tuning Language Models from Human Preferences

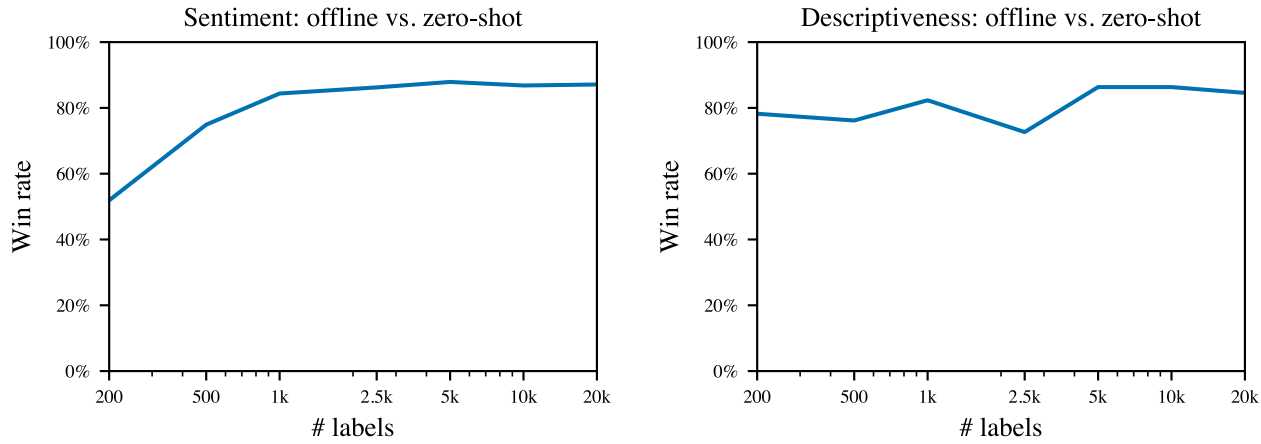


Figure 4: Human evaluations comparing the zero-shot model with offline fine-tuned models using varying amounts of human data. We report how often the fine-tuned model is preferred by a majority of 3 labelers. We omit error bars because we lack an estimate of the largest source of variance (randomness across training runs).

	Sentiment			Descriptiveness		
5k offline vs. zero-shot	88%		12%	86%		14%
5k offline vs. mock	77%		23%	—		
5k offline vs. 20k offline	48%		52%	47%		53%
5k offline vs. 5k online	50%		50%	48%		52%

Table 1: Human evaluations for the sentiment and descriptiveness tasks. We sample 1024 excerpts from the BookCorpus test set and report how often each model’s continuations were preferred, as judged by a majority of 3 labelers.

$\text{KL}(\pi, \rho)$ between the learned policy π and the language model ρ , the optimal policy has the form:

$$\pi_{\text{opt}}(y|x) \propto \rho(y|x)e^{r_s(x,y)/\beta}$$

We approximate the reward of this policy for given x and β by sampling a large number of continuations from $\rho(y|x)$ and reweighting them by $e^{r_s(x,y)/\beta}$. Figure 3 compares the reward obtained by our policies to the estimated optimal reward across a range of KL values. There is a significant gap from optimality after training the policy on 2M continuations—the number used in our main experiments—though it is largely closed with more training. Our policies continue to receive higher rewards for larger KL divergences, where we cannot afford to approximate π_{opt} by sampling.

3.1.2. HUMAN EVALUATIONS OF CONTINUATIONS

We apply our method to two continuation tasks defined by human judgments:

Sentiment: Humans are asked to reward “positive and happy” continuations.

Descriptiveness: Humans are asked to reward “vividly descriptive” continuations.

The human labelers are presented with a BookCorpus excerpt and four possible continuations; they are asked to select the best continuation. Full instructions for labelers are provided in appendix A (although labelers also learned from ~ 50 example comparisons labeled by the authors and so the instructions do not completely define the task).

To make the labeling task more natural, we select excerpts that start and end with a period. When sampling continuations that will be presented to humans, we use rejection sampling to ensure there is a period between tokens 16 and 24 and then truncate at that period.⁴ During the RL fine-tuning, we penalize continuations that don’t have such a period by giving them a fixed reward of -1 .

We dynamically adjusted β to obtain a KL divergence of 6 nats for descriptiveness and 10 nats for sentiment (section 2.2).

We trained a range of models using different amounts of feedback, testing both offline data collection where humans

⁴This is a crude approximation for “end of sentence.” We chose it because it is easy to integrate into the RL loop, and even a crude approximation is sufficient for the intended purpose of making the human evaluation task somewhat easier.

基于人类偏好对语言模型进行微调

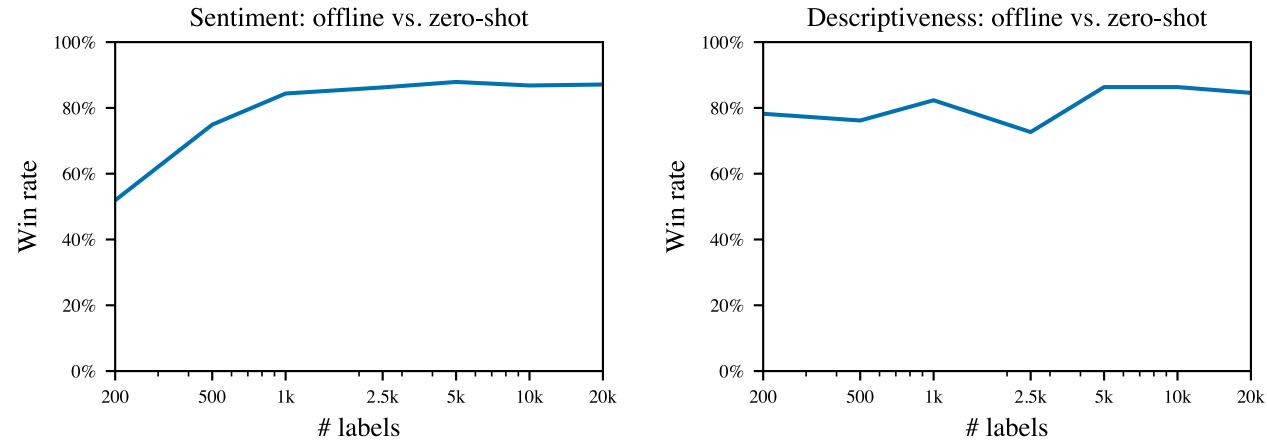


图4展示了使用不同量级的人类数据时，人类评估结果对零样本模型与离线微调后的模型的对比情况。我们统计了在多少比例的3名标注员中，微调后的模型更受多数人青睐。由于无法估算方差的最大来源（即训练轮次中的随机性），因此此处未标注误差条。

	情感倾向			描述性		
5k离线测试与 零样本学习	88%		12%	86%		14%
5k离线测试与 模拟任务	77%		23%	—		
5k离线测试与20k离线测试	48%		52%	47%		53%
5k离线测试与5k在线模式测试	50%		50%	48%		52%

表1：针对情感倾向与描述性分析任务的人类评估结果。我们从BookCorpus测试集中随机抽取了1024段文本，统计在3名标注员的多数判定下，各模型的续写结果被优先选择的频率。

在所学策略 π 与语言模型 ρ 之间的 $\text{KL}(\pi, \rho)$ 作用下，最优策略的形式为：

$$\pi_{\text{opt}}(y|x) \propto \rho(y|x)e^{r_s(x,y)/\beta}$$

我们通过与从 $\rho(y|x)$ 中抽取大量续写结果，并利用 $e^{r_s(x,y)/\beta}$ 对它们进行权重调整，以此来近似计算给定 x 和 β 条件下该策略的奖励值。图3展示了我们的策略所获得的奖励值与不同KL散度数值下估算出的最优奖励值之间的对比。虽然在主要实验中使用了200万条续写数据对策略进行训练，其结果与最优值仍存在显著差距，但随着训练数据量的增加这一差距已大幅缩小。此外，当KL散度较大时，我们的策略依然会获得更高的奖励，因为在这样的情况下我们无法再通过抽样方式来进行近似计算 π_{opt} 。

3.1.2. 人类对续写的评估

我们将该方法应用于两项由人类判断定义的续写任务：

情感倾向：任务要求参与者为“积极且愉悦”的续写结果给予奖励。

描述性：研究们需要为那些“描述极为生动”的续写结果给予奖励。

这些人工标注员会看到一段来自BookCorpus语料库的文本以及四种可能的续写内容，随后需从中选出最佳选项。关于标注工作的完整说明可见附录A（不过标注员们也会参考 ~ 50 研究者们标注过的示例对比，因此这些说明并不能完全界定该任务的全部要求）。为了让标注任务更符合实际使用场景，我们选取了以句号开头和结尾的文本片段。在为人类展示续写结果时，我们会采用拒绝采样方法，确保第16个与第24个标记之间存在一个句号，随后在该处截断文本。⁴ 在通过强化学习进行微调的过程中，对于没有此类句号的续写结果，我们会通过给予固定奖励 -1 的方式对其予以惩罚。

我们会动态调整 β ，使得描述性的KL散度为6纳特，情感倾向的KL散度为10纳特（第2.2节）。

我们使用不同量的反馈训练了多种模型，并测试了在离线数据收集环境中由人类参与的场景。

⁴这仅是“句子结束”概念的一种粗略近似。我们选择这种方式是因为它易于整合到强化学习循环中，而且即便只是粗略近似，也能达到让人类评估任务相对轻松的预期目的。

Fine-Tuning Language Models from Human Preferences			
context	Pearl thought to herself that what they were about to do was exactly the sort of thing that they could do to help the villagers. They were all terrified of these guys. At the police station the three walked up to the counter behind which was a senior constable studying some papers.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	"Hello, I'm Pearl and this is my friend, Mike," said Pearl.	"May we speak to the police officer, sir?" asked the one in charge.	'Hello, can I help you?' 'Yes, we're the same people that the people were talking about.
5k offline fine-tune	He turned to them with a smile. "Good afternoon, ladies. I'm Detective Inspector Jones.	The constable stood up and smiled as he saw them, obviously pleased to see them.	He smiled at them and waved them in, his eyes twinkling as he listened to their tales.

Table 2: Three random ($T = 0.7$) continuations for our sentiment continuation task. Chosen from appendix [table 11](#); see appendix for more.

context	"I do not know if it was Viking related, but it could have been." "Really?" Ailia said. Is it safe to be traveling here then? Ailia looked behind her to make sure they weren't being followed.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	There were no signs of anyone. "It is safe enough," Ailios said.	"Because I have a friend that is in the area and he will be coming with us.	It was hard to see that far. "I do not like that word.
5k offline fine-tune	Kaya crouched low, her eyes wide in the moonlight. Her body was tense.	She put her hand on the sword strapped to her back, and then pulled it out.	She strode out the door and walked down the street, her nose wrinkled in disapproval.

Table 3: Three random ($T = 0.7$) continuations for our descriptiveness continuation task. Chosen from appendix [table 12](#); see appendix for more.

rate only the initial language model’s continuation, and on-line data collection where humans continuously rate the current policy’s continuations ([section 2.3](#)). We then compared these different policies to each other and to the zero-shot performance of the original language model. The results are shown in [fig. 4](#) and [table 1](#). Each model comparison is based on 1024 four-way continuation comparisons, two from each of the models being compared, each rated by 3 humans.

For these continuation tasks, offline and online data collection give similar performance. We find that very little human data is required for fine-tuning: performance with 5k, 10k, and 20k reward model training samples is similar, degrading only for less than 5k samples.⁵ The model trained using the review sentiment classifier from [section 3.1.1](#) does poorly relative to models optimized using human preference: in 77% of contexts, labelers preferred the output of the model trained with real human feedback.

⁵The descriptiveness policy trained with 2.5k samples performed poorly, but we believe this is due to randomness in RL.

3.2. Summarization

We also applied our method to two summarization tasks: the CNN/Daily Mail dataset of [Hermann et al. \(2015\)](#) and the TL;DR dataset of [Völske et al. \(2017\)](#). We sample articles or Reddit posts, truncate to 500 tokens, add a "`\n\nTL;DR:`" suffix (and for CNN/Daily Mail, a "`Article:\n\n`" prefix) and let the policy respond with up to 75 tokens. We set the temperature of the pretrained model to $T = 0.5$ for CNN/Daily Mail and $T = 0.7$ for TL;DR. To make the task more natural for humans, we ensure articles consist of whole sentences by truncating to the last newline character. When sampling summaries that will be shown to a human, we use rejection sampling to ensure there is a newline between tokens 55 and 75 and truncate at that newline. During RL fine-tuning, we penalize summaries that don’t have such a newline by giving them a fixed score of -1. For CNN/Daily Mail we used a fixed KL coefficient $\beta = 0.1$; for TL;DR we used $\beta = 0.03$.

For RL fine-tuning, we trained online data collection mod-

基于人类偏好对语言模型进行微调			
上下文	珀尔在心里想，他们即将要做的事正是那种能够帮助村民的事情。他们全都非常害怕那些人。 在警察局里，这三个人走向柜台，柜台后面坐着一名正在研究文件的高级警官。 在警察局里，三人走向柜台，柜台后面坐着一名资深警官。 一名警员正在研读一些论文。		
	续写1	续写2	续写3
零样本学习	"你好，我是Pearl，这是我的朋友Mike，" Pearl说道。	"请问我们可以和警察谈谈吗？" "是警官吗？"里面的那名警察问道。费用。	'您好，有什么可以帮您的吗？' '是的，我们就是之前那些人。' "也就是刚才正在交谈的那群人。"
5k 离线模式 微调	他带着一丝微笑走向他们。 "下午好，" 女士们，我是警探英斯佩克-托尔·琼斯。	那名警官站起身来，看到他们后露出了笑容，显然非常高兴见到他们。	关于·他朝他们微笑并挥手致意。他眼眸闪亮地将他们迎了进来。随后他便认真聆听了他们的故事。

表2：我们用于情感倾向续写任务的三个随机 ($T = 0.7$) 续写结果。这些样本选自附录中的[表11](#)；更多详情请参见附录。

上下文	"我不知道这是否与维京人有关，但也有可能就是。" "真的吗？"艾莉亚问道，"那在这里旅行安全吗？"说罢她向身后望去。 要确保没有人在跟踪他们。		
	续写1	续写2	续写3
零样本学习	没有任何跟踪的迹象。 one. "这已经足够安全了，" 艾利奥斯说道。 said.	"因为我有个朋友在附近，"他也会和我们一同前往。他就在那个区域，他会跟我们一起来的，他会和我们一起过来。	那么远的地方实在难以看清楚。『我……』 我不喜欢那个词。
5k 离线模式 微调	卡雅蹲低身子，双眼紧盯着前方。她的身体十分紧绷。她的身体处于紧张状态。	她将手放在……上。她将剑背在背上。随后便将其抽出。	她大步走出了门。接着沿街道走去。她的鼻子因不满而皱了起来。 这无疑是 一种证明。

表3：针对我们的描述性续写任务生成的三个随机 ($T = 0.7$) 续写结果。这些结果选自附录[表12](#)；更多详情请参见附录。

仅评估初始语言模型的续写结果，[第2.3节](#)。随后我们将这些不同的策略相互比较，同时也将其与原始语言模型的零样本学习性能进行了对比。相关结果展示在[图4](#)以及[表1](#)中。每项模型对比都是基于1024次四向续写结果对比得出的，即从被对比的每款模型中各选取2次对比，再由3名人类对其进行评分。

对于这些续写任务而言，离线模式与在线模式下的数据收集所带来的性能表现相当。我们发现，微调过程几乎不需要太多人类数据：使用5千、10千以及20千个奖励模型训练样本时的性能差异很小，只有当样本数量少于5千时性能才会出现下降。⁵ 使用[3.1.1节](#)中介绍的评论情感分类器训练得到的模型，与那些基于人类偏好优化的模型相比性能较差：在77%的上文中，标注员更倾向于选择经过真实人类反馈训练所得出的模型输出。

⁵使用2500个样本训练得到的描述性策略表现不佳，但我们认为这是由于强化学习过程中存在的随机性所致。

3.2. 摘要生成

我们还把该方法应用到了两项摘要生成任务中：分别是 [Hermann等人（2015年）](#) 所构建的CNN/每日邮报数据集，以及[Völske等人（2017年）](#) 开发的TL;DR数据集。我们会选取文章或Reddit帖子，将其长度截断至500个标记，随后添加 “`\n\nTL;DR:`” 这样的后缀（而对于CNN/每日邮报数据集，则会添加 “`Article:\n\n`” 这样的前缀），再让策略生成最多75个标记长的摘要。我们会为预训练模型设置不同的温度参数，CNN/每日邮报数据集使用 $T = 0.5$ 该数值，而TL;DR数据集则使用 $T = 0.7$ 该数值。为了让这项任务对人类来说更自然，我们会通过截断到最后一个换行符的方式，确保文章的内容均由完整的句子构成。在为人类展示而生成摘要时，我们会采用拒绝采样方法，保证第55个和第75个标记之间存在一个换行符，并在该位置进行截断。在强化学习微调过程中，对于没有此类换行符的摘要，我们会给予其-1的固定分数作为惩罚。针对CNN/每日邮报数据集，我们使用了固定的KL系数 $\beta = 0.1$ ；而针对TL;DR数据集，则使用了 $\beta = 0.03$ 。

在强化学习微调中，我们对在线数据收集模块进行了训练。

Fine-Tuning Language Models from Human Preferences								
	TL;DR				CNN/Daily Mail			
	R-1	R-2	R-L	R-AVG	R-1	R-2	R-L	R-AVG
SOTA	22*	5*	17*	14.7*	41.22	18.68	38.34	32.75
lead-3	17.435	3.243	14.575	11.751	40.379	17.658	36.618	31.552
zero-shot	15.862	2.325	13.518	10.568	28.406	8.321	25.175	20.634
supervised baseline	17.535	3.124	14.969	11.877	39.525	16.992	36.728	31.082
supervised + 60k fine-tune	18.434	3.542	15.457	12.478	40.093	17.611	37.104	31.603
60k fine-tune	16.800	2.884	14.011	11.232	37.385	15.478	33.330	28.731
30k fine-tune	16.410	2.920	13.653	10.994	35.581	13.662	31.734	26.992
15k fine-tune	15.275	2.240	12.872	10.129	38.466	15.960	34.468	29.631
60k offline fine-tune	16.632	2.699	13.984	11.105	33.860	12.850	30.018	25.576

Table 4: ROUGE evaluations of summarization models. For all models (excluding the lead-3 baselines), we sample with temperature 0.7 for TL;DR and 0.5 for CNN/Daily Mail. We use the CNN/DM test set, but our own validation set for TL;DR. CNN/Daily Mail SOTA is from [Gehrmann et al. \(2018\)](#). * TL;DR SOTA is from [Gehrmann et al. \(2019\)](#), but the numbers are not comparable as we lack test set access and the TL;DR leaderboard uses an unofficial implementation of ROUGE.

	TL;DR			CNN/Daily Mail		
60k fine-tuned vs. zero-shot	96%		4%	91%		9%
60k fine-tuned vs. supervised	97%		3%	80%		20%
60k fine-tuned vs. lead-3	45%		55%	40%		60%
60k fine-tuned vs. supervised + 60k fine-tuned	80%		20%	74%		26%
60k fine-tuned vs. 30k fine-tuned	40%		60%	62%		38%
60k fine-tuned vs. 15k fine-tuned	79%		21%	47%		53%
60k fine-tuned vs. 60k offline fine-tuned	64%		36%	65%		35%
60k fine-tuned vs. reference summaries	96%		4%	84%		16%
lead-3 vs. supervised	97%		3%	89%		11%
lead-3 vs. reference summaries	97%		3%	89%		11%
lead-3 vs. supervised + 60k fine-tuned	75%		25%	85%		15%

Table 5: Human evaluation of summarization models. For each pair of models and each dataset, we sample 1024 articles from the test set, generate a summary from each model, and ask 3 humans to pick the best summary using the same instructions as in training. The model chosen by a majority of the humans wins on that article. We report the fraction of articles that each model wins. For all models, we sample with temperature 0.7 for TL;DR and 0.5 for CNN/DM.

els with 15k, 30k, and 60k human labels, and an offline data collection ablation with 60k labels. We also show zero-shot performance of the pretrained model, a supervised fine-tuned baseline using the same pretrained model as starting point ([section 2.2](#)), and a lead-3 baseline which copies the first three sentences of the context. We truncate lead-3 at a period in the same way we truncate generated summaries, so occasionally it is 2 sentences. Finally, we combine supervised and RL fine-tuning: performing human RL fine-tuning starting with the supervised fine-tuned model. The purely RL fine-tuned models use contexts from the datasets during training but ignore the reference summaries; the supervised and supervised+RL models use both contexts and summaries.

We report two sets of numerical results: human evaluations between pairs of models ([table 5](#)) and ROUGE results on the test set of CNN/Daily Mail and our validation set of TL;DR ([table 4](#)). ROUGE results suggest that online data collection is important for best performance, in contrast to our stylistic continuation tasks. At a fixed number of labels, online tends to be better than offline, with a 3 point R-AVG gain on CNN/DM at 60k labels.⁶ On both datasets we see significant returns to data volume up to 60k human labels (though the trend is less clear for human evaluation). On both datasets, supervised + RL fine-tuning is best, and indeed pure RL fine-tuning is worse than the supervised baseline according to

⁶That said, different training runs have considerable variation and it is expensive to run multiple seeds with humans, so it is possible that this gap is largely noise.

基于人类偏好对语言模型进行微调								
	简而言之				CNN/每日邮报			
	R-1	R-2	R-L	R-AVG指标	R-1	R-2	R-L	R-AVG指标
SOTA	22*	5*	17*	14.7*	41.22	18.68	38.34	32.75
Lead-3模型	17.435	3.243	14.575	11.751	40.379	17.658	36.618	31.552
零样本学习	15.862	2.325	13.518	10.568	28.406	8.321	25.175	20.634
有监督基准模型	17.535	3.124	14.969	11.877	39.525	16.992	36.728	31.082
经过 + 60千次样本的监督学习微调	18.434	3.542	15.457	12.478	40.093	17.611	37.104	31.603
6万次微调	16.800	2.884	14.011	11.232	37.385	15.478	33.330	28.731
3万样本微调	16.410	2.920	13.653	10.994	35.581	13.662	31.734	26.992
1.5万样本微调	15.275	2.240	12.872	10.129	38.466	15.960	34.468	29.631
6万样本离线微调	16.632	2.699	13.984	11.105	33.860	12.850	30.018	25.576

表4：摘要生成模型的ROUGE分数评估结果。对于所有模型（lead-3基线模型除外），在处理TL;DR任务时我们采用温度参数为0.7的采样策略，而在处理CNN/每日邮报任务时则采用温度参数为0.5的策略。评估所使用的测试集为CNN/每日邮报测试集，而TL;DR任务的验证集则由我们自行构建。CNN/每日邮报任务目前的当前最佳性能数据出自[Gehrmann等人\(2018年\)](#)。* TL;DR任务的当前最佳性能数据同样出自[Gehrmann等人\(2019年\)](#)，但由于我们无法获取对应的测试集，且TL;DR任务的排行榜使用的是非官方版本的ROUGE评分标准，因此这两个数值并不具备可比性。

	TL;DR			CNN/每日邮报		
经过6万次微调的模型与零样本学习	96%		4%	91%		9%
经过6万次微调的模型versus 监督学习	97%		3%	80%		20%
经过6万次微调的模型versus Lead-3模型	45%		55%	40%		60%
经过6万次微调的模型versus 监督学习 + 60以及经过1.5万参数微调模型	80%		20%	74%		26%
经过6万次微调的模型versus 3万样本微调	40%		60%	62%		38%
经过6万次微调的模型versus 1.5万参数微调模型	79%		21%	47%		53%
经过6万次微调的模型versus 6万样本离线微调	64%		36%	65%		35%
经过6万次微调的模型与参考摘要	96%		4%	84%		16%
Lead-3模型与监督学习方法	97%		3%	89%		11%
Lead-3模型与参考摘要	97%		3%	89%		11%
Lead-3模型与监督学习方法 + 60万次微调的模型	75%		25%	85%		15%

表5：摘要生成模型的人类评估结果。对于每一组模型和每一个数据集，我们会从测试集中随机抽取1024篇文章，让每款模型分别为这些文章生成摘要，随后请3名受试者按照训练时相同的指示选出最佳摘要。获得多数人认可的模型即为该篇文章的胜出者，我们最终会统计每款模型获胜的文章比例。所有模型的采样过程中，TL;DR类型的模型使用温度参数为0.7，而CNN/DM类型的模型则使用0.5。

包含15千、30千和60千条人类标注的ELS，以及一项使用60千条标注进行的离线数据收集消融实验。我们还展示了预训练模型的零样本学习性能、以同一预训练模型为起点的监督学习微调基线模型（[第2.2节](#)），以及复制上下文前三个句子的Lead-3基线模型。我们与处理生成摘要时相同的方式，在句号处对Lead-3模型进行截断，因此有时它的句子数仅为2个。最后，我们结合了监督学习微调和强化学习微调：以经过监督学习微调的模型为基础，进一步进行人类引导的强化学习微调。纯粹通过强化学习微调得到的模型在训练过程中会使用数据集中的上下文，但会忽略参考摘要；而那些经过监督学习的模型+则会同时使用上下文和摘要。

我们报告了两组数值结果：模型对之间的人类评估结果（[表5](#)）以及CNN/每日邮报测试集和TL;DR验证集上的ROUGE分数结果（[表4](#)）。ROUGE分数表明，与我们的风格化文本续写任务不同，要想获得最佳性能，进行在线数据收集十分重要。在标签数量固定的情况下，在线模式的性能通常优于离线模式，例如在CNN/DM数据集且标签数为60k时，其R-AVG指标可提升3分。⁶在这两个数据集上，我们都观察到当人类标注数量达到60k之前，数据量越大带来的性能提升越显著（不过人类评估方面的这一趋势则不那么明显）。在两个数据集中，采用+监督学习下的强化学习微调效果最为出色，事实上，根据相关研究，单纯的强化学习微调表现甚至不如监督基准模型

⁶不过，不同的训练轮次之间存在较大差异，而且让多名受试者参与多轮测试的成本相当高昂，因此这种差距很可能是由随机噪声造成的。

Fine-Tuning Language Models from Human Preferences	
reference summary	Solar plane attempting to be first to circumnavigate world without using fuel is stuck in China. Solar Impulse 2 attempts to prove the power of renewable energy.
zero-shot	The plane has been grounded in China for two weeks because of bad weather.
60k fine-tune	The Solar Impulse 2, the experimental plane attempting to fly around the world without using a drop of fuel, has been grounded by the weather in China. What was supposed to be an overnight pit stop in the southwestern city of Chongqing has now stretched into a two-and-a-half week stay.
supervised	Solar Impulse 2 has been grounded by the weather in China. The plane took off from Abu Dhabi on March 9. The plane is trying to prove the power of renewable energy.
supervised + 60k fine-tune	Solar Impulse 2 has been grounded in China for two-and-a-half weeks. The plane is attempting to fly around the world without using a drop of fuel. The team, which includes Bertrand Piccard, is taking turns flying the single-seater. The plane took off from Abu Dhabi on March 9 and has successfully flown through Oman.

Table 6: Random ($T = 0.5$) summaries for our CNN/DM summarization task, on the same context. Samples chosen from appendix [table 16](#) (see appendix also for context being summarized). The 60k fine-tune model copies from the source article.

ROUGE in all cases (though the supervised baseline uses the full supervised training dataset, which is much larger than 60k samples). Lead-3 is hard to beat: it is the best model for R-1 and R-2 on CNN/Daily Mail, and only supervised + RL fine-tuning beats it otherwise.

But our goal is optimizing reward defined by humans, not ROUGE. [Table 5](#) shows pairwise comparisons between different model pairs according to human labelers, using 1024 samples with majority vote of 3 labelers per sample. Here the picture is different, though also significantly noisier. Our online trained, 60k label model reliably beats both the zero-shot and supervised baselines, and even beats the combined supervised + RL fine-tuned model. Online training remains important, but the situation w.r.t. data volume is less clear and likely contaminated by noise: the 60k TL;DR model beats the 30k model only 40% of the time, for example. More worrisome, the 60k online model beats the human ground truth 96% of the time for TL;DR and 84% of the time for CNN/Daily Mail.

What is going on? As we show in the next section, our 60k RL fine-tuned model is almost entirely extractive (despite lacking any explicit extractive architectural component): it mostly copies whole sentences from the context, but varies which sentences are copied.

3.2.1. WHAT OUR MODELS COPY

Much previous work in summarization has focused on explicit copying mechanisms, including the pointer network-based architecture of [See et al. \(2017\)](#) and the two-phase mask and paraphrase approach of [Gehrmann et al. \(2018\)](#). The goal is to take advantage of copying (which is of fundamental importance to the task of summarization) without only copying—to be abstractive rather than extractive.

[Figures 5](#) and [6](#) show the fractions of n -grams and sentences generated by our models which are novel and repeated, respectively. From the novelty stats, we see that our RL fine-tuning consistently causes models to copy more. In particular, our 60k RL fine-tuned models are almost entirely extractive: they copy whole sentences 71% of the time for TL;DR and 98% of the time for CNN/Daily Mail. Applying RL fine-tuning starting from the supervised fine-tuned model copies much less: 6% and 30% for TL;DR and CNN/Daily Mail. Although we do not use explicit coverage metrics as in [See et al. \(2017\)](#); [Gehrmann et al. \(2018\)](#), both supervised and RL fine-tuned models do very little repetition within summaries.

While the purely RL fine-tuned models mostly copy, they vary where they copy from. [Figure 7](#) illustrates this via the position of the longest common subsequence between context and summary. To understand when the model chooses to copy from the exact beginning, we identify common preambles in articles such that we would expect copying to be a poor strategy. [Table 7](#) shows that these preambles are copied much less often than in the immediate beginnings of other articles, giving evidence that our models are smart about when to copy. However, we cannot establish that our reward model is smart beyond rewarding copying, as the zero-shot model also skips preambles.

Since combining supervised fine-tuning and RL fine-tuning gives the best ROUGE scores and and is also more abstractive, why not use it? Unfortunately there is an advantage to pure copying shown in [table 8](#): it makes it easy for the model to tell the truth. The models that copy the most, 60k RL fine-tuned, is 90% and 95% accurate on TL;DR and CNN/Daily Mail; lifting whole sentences from the article usually leaves them true. The supervised fine-tuned and combined supervised+RL fine-tuned models are accurate

基于人类偏好对语言模型进行微调	
参考资料摘要	那架试图成为首架无需燃料即可环游世界的太阳能飞机目前陷入了困境。在中国，太阳脉冲2号正努力证明可再生能源的巨大潜力。
零样本学习	由于恶劣天气，这架飞机已在中国停飞两周。
6万次微调	试图在不使用一滴燃料的情况下环游世界的实验性飞机——太阳脉冲2号，也因中国境内的恶劣天气而被迫停飞。原本计划在该西南城市进行的一夜停留，如今已延长至两周半之久。 本应是在西南部城市进行的一次过夜技术调整在重庆的行程现在已经延长到了两周半。 由于中国境内的恶劣天气，阳光脉冲2号不得不停飞。该飞机是从阿布扎比于3月9日起飞的，此次飞行旨在证明可再生能源的巨大潜力。
监督学习	它目前在中国停飞已有两个半星期，仍在努力完成它的使命。
监督学习 + 6万次微调	阳光脉冲2号目前仍在中国停留，等待天气状况好转以便继续它的环球飞行之旅。正试图在不使用一滴燃料的情况下环游世界。该团队由贝特朗·皮卡尔等人组成，轮流驾驶这架单座飞机。这架飞机于3月9日从阿布扎比起飞，目前已成功飞越阿曼境内。于3月9日从阿布扎比出发，已顺利飞越阿曼。

表6：在相同上下文中，针对我们的CNN/DM摘要生成任务生成的随机（ $T = 0.5$ ）摘要。这些样本选自附录[表16](#)（有关被摘要的上下文信息，请另见附录）。该6万次微调后的模型是复制自原始文章的。

在所有案例中都是如此，尽管有监督基准模型使用了规模远大于60k样本的完整监督训练数据集。Lead-3模型的表现极难被超越：它在CNN/每日邮报数据集上的R-1和R-2评估指标中均为最优模型，只有在采用有监督+强化学习微调的方式时，才有其他模型能够超越它。

“总结而言，虽然在线训练依旧重要，但由于数据量的限制，其效果并不总那么理想，而且数据中往往存在干扰因素。比如，6万样本TL;DR模型仅有40%的时间能胜过3万样本模型。更为严重的是，该模型在生成TL;DR摘要时96%的时间能超越人工真实标签，而在生成CNN/每日邮报摘要时这一比例仅为84%。”

[图5](#)和[图6](#)分别展示了我们的模型生成的 nn 元词组与句子中属于新颖内容与重复内容的占比。从新颖性统计结果可以看出，强化学习微调会使得模型更频繁地进行复制行为。尤其是经过60k次强化学习微调的模型几乎完全采用信息提取模式：在生成TL;DR摘要时，有71%的时间会复制整句；而在生成CNN/每日邮报格式的摘要时，这一比例高达98%。而如果在已通过监督学习微调的模型基础上再施加强化学习微调，复制行为则会大幅减少，生成TL;DR摘要时的复制比例为6%，生成CNN/每日邮报格式摘要时的复制比例为30%。尽管我们并未像[相关文献（2017年）](#)以及[Gehrmann等人（2018年）](#)那样使用明确的覆盖率指标，但无论是通过监督学习微调还是强化学习微调的模型，其生成的摘要中重复内容都非常少。

究竟发生了什么？正如我们在下一节中所展示的，尽管该模型并未采用任何专门的提取结构组件，但经过60k次强化学习微调后的版本几乎完全依赖复制行为——它主要从上下文中复制整句内容，不过所复制的句子会有所不同。

3.2.1. 我们的模型复制了什么

此前在摘要生成领域的诸多研究都侧重于显式的复制机制，其中包括基于指针网络的架构[参见相关文献（2017年）](#)，以及[Gehrmann等人](#)提出的两阶段掩码与改写方法（2018年）。这些研究的目的是充分利用复制行为——该行为对摘要生成任务而言具有根本性重要性——但同时避免单纯的复制，从而实现抽象化表达而非简单的信息提取。

[图5](#)和[图6](#)分别展示了我们的模型生成的 nn 元词组与句子中属于新颖内容与重复内容的占比。从新颖性统计结果可以看出，强化学习微调会使得模型更频繁地进行复制行为。尤其是经过60k次强化学习微调的模型几乎完全采用信息提取模式：在生成TL;DR摘要时，有71%的时间会复制整句；而在生成CNN/每日邮报格式的摘要时，这一比例高达98%。而如果在已通过监督学习微调的模型基础上再施加强化学习微调，复制行为则会大幅减少，生成TL;DR摘要时的复制比例为6%，生成CNN/每日邮报格式摘要时的复制比例为30%。尽管我们并未像[相关文献（2017年）](#)以及[Gehrmann等人（2018年）](#)那样使用明确的覆盖率指标，但无论是通过监督学习微调还是强化学习微调的模型，其生成的摘要中重复内容都非常少。

虽然纯通过强化学习微调得到的模型大多会进行复制，但它们复制的内容位置各不相同。[图7](#)通过展示上下文与摘要之间的最长公共子序列的位置，直观体现了这一现象。为了解释模型为何有时会选择从开头直接复制内容，我们首先识别出文章中那些通常不适合通过复制来生成摘要的常见开头部分。[表8](#)显示，这些开头部分的复制频率远低于其他文章的开头部分，这证明了我们的模型在判断何时复制内容时相当明智。不过，由于零样本模型也会跳过这类开头部分，我们仍无法确定其奖励模型除了奖励复制行为之外还具备其他智能判断能力。

既然结合监督学习微调与强化学习微调不仅能获得更高的ROUGE分数，还能让模型的表现更具抽象性，为何不直接采用这种方案呢？遗憾的是，如[表8](#)所示，纯粹的复制行为也有其优势：它能让模型更易于输出真实内容。那些复制频率最高的60k次强化学习微调模型，在TL;DR和CNN/每日邮报这两个数据集上的准确率分别达到了90%和95%；因为直接复制整句内容通常能够保留原文的真实性。而经过监督学习微调以及两种方法结合训练的模型+强化学习微调模型也具有较高的准确率

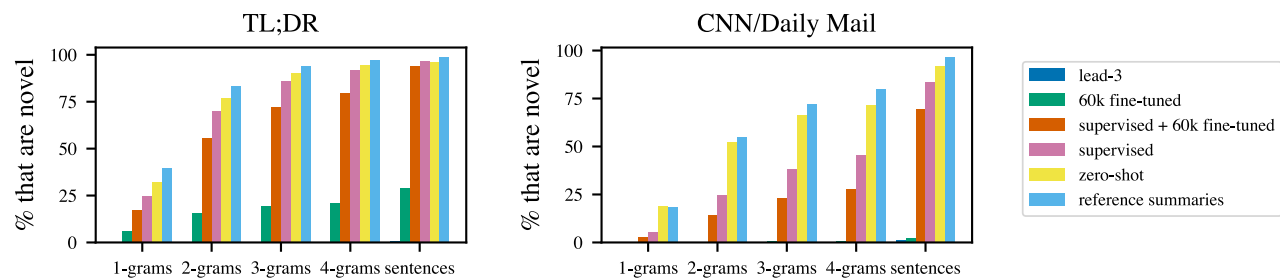


Figure 5: Percent of n -grams and sentences in summaries that do not appear in the source (compare to figure 6 in [See et al. \(2017\)](#)). n -grams are consecutive sequences of words in a single sentence in a summary, and they count as novel if they do not appear consecutively in the article. We ignore punctuation and capitalization.

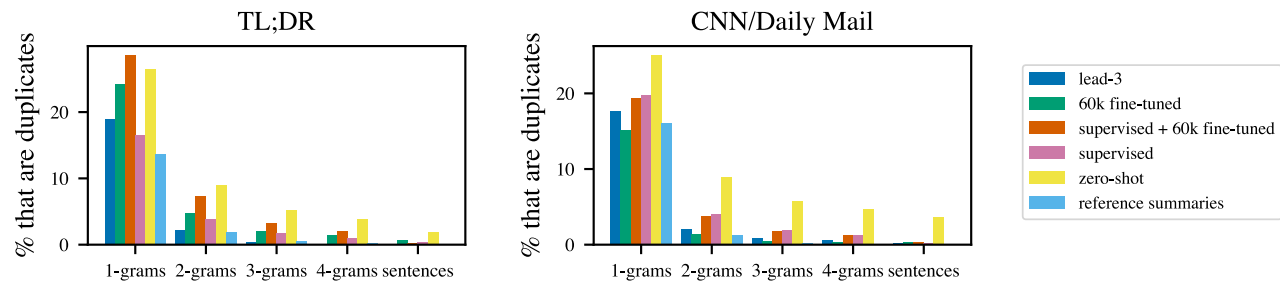


Figure 6: Percent of n -grams and sentences in summaries that appear multiple times in the summary (compare to figure 4 in [See et al. \(2017\)](#)).



Figure 7: Variation in where the models copy from, illustrated by the location of the longest common subsequence of bigrams between context article/post (left) and summary (right) for 256 randomly chosen contexts. Document lengths are shown in gray, with bigrams highlighted (with color depending on positions in contexts). Here, we picture the 60k fine-tuned models, which do the most copying.

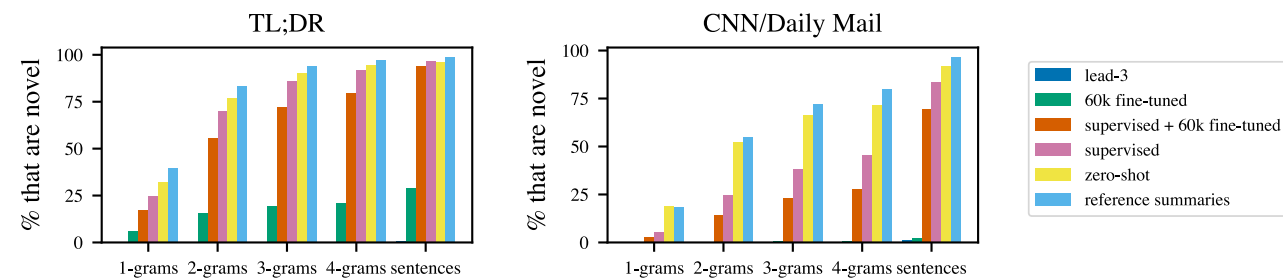


图5: n 摘要中出现但原始文本中不存在的二元组与句子所占比例（可参照[参见相关研究\(2017年\)](#)中的图6）。 n 二元组是指摘要中单个句子内的连续词序列，若在原始文章中并非连续出现，则被视为新内容；同时我们会忽略标点符号和大小写的影响。

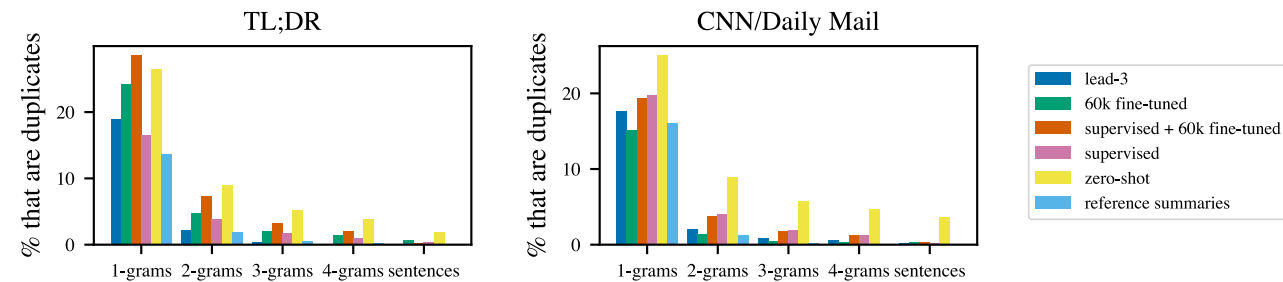


图6: n 摘要中出现多次的二元组与句子所占比例（可参照[参见相关研究\(2017年\)](#)中的图4。



图7: 展示了256个随机选取的上下文文章/帖子与对应摘要之间二元组的最长公共子序列位置，进而反映了模型复制内容的差异。图中灰色部分表示文档长度，而标有颜色的二元组则根据其在上下文中的位置有所不同。此处展示的是经过6万次微调的模型，这类模型的复制行为最为频繁。

Fine-Tuning Language Models from Human Preferences				
	TL;DR		CNN/Daily Mail	
	all	preamble	all	preamble
zero-shot	1.3%	0%	10.4%	1.0%
60k fine-tuned	28.3%	0.2%	77.6%	1.4%
supervised	1.5%	0%	9.4%	0%
supervised + 60k fine-tuned	7.9%	0%	16.6%	0%
reference summaries	0.6%	0%	5.1%	0%
total articles	30000	3762	13368	297

Table 7: How often different models copy the first 3 words of the article as the first 3 words of the summary, on the validation sets. We additionally consider the subset of posts/articles with preambles. On TL;DR, we used posts which begin with one of ‘hi’, ‘hello’, ‘hey’, ‘ok’, ‘okay’, or ‘so’. For CNN/Daily Mail, we used articles with a colon within the first 3 words, such as “Winner: Simon Wood took home the TV crown [...]” and “Fully charged: The new scheme will let EE customers pick up free portable chargers [...]”.

	TL;DR	CNN/Daily Mail
zero-shot	6/30	6/30
60k fine-tuned	26/30	29/30
supervised	8/30	19/30
supervised + 60k fine-tuned	11/30	20/30

Table 8: Frequency with which generated summaries are accurate, in the sense of only making statements supported by the context, as judged by the authors on 30 articles from each dataset. The 60k fine-tuned model achieves high accuracy via copying; the supervised and supervised + 60k fine-tuned models are more abstractive but at significant cost to accuracy.

at most 70% of the time: they paraphrase but paraphrase badly, often swapping names from the context or mixing together multiple sentences in invalid ways. Zero-shot is the most novel, but is accurate only 20% of the time. Similarly, [Kryściński et al. \(2019\)](#) found that 30% of samples from the supervised summarization models they tested contained inconsistencies, and [Khandelwal et al. \(2019\)](#) found that their pretrained encoder-decoder model “hallucinates facts...which are topical but never appear in the source”.

There are at least two ways of interpreting these results. The first is that copying is the easiest way to be accurate. The labelers were told to penalize inaccuracy and redundancy, but were not told to penalize copying. The zero-shot model copies some of the time, and when it copied it was accurate, so this behavior was reinforced. The result is a model that “degenerated to copying”, but at least does not lie.

However, this does not explain why both our model and lead-3 are strongly preferred by the labelers to the human reference summaries ([table 5](#)). This reveals a mismatch between the notion of quality we wanted our model to learn, and what the humans labelers actually evaluated. Checking for copying is very easy, so labelers who check primarily for copying can work quickly. Since the online data collection setting made quality control more difficult, we failed to detect and penalize this behavior.

4. Challenges

We conclude with a few lessons and directions we plan to consider in future reward learning work.

4.1. Online data collection is hard

Online data collection was necessary to achieve the best results on summarization. However, fully online data collection—where each label comes from an up-to-date version of the policy which has already learned from almost all previous labels—had major disadvantages:

- Software complexity:** Our online system interleaves data gathering, reward model training, and RL fine-tuning. The resulting distributed system was significantly more complicated than if each task was kept separate, slowing the software development process. Moreover, a bug in any one of the tasks would break the entire training process.
- Machine learning complexity:** Online experiments were difficult to debug, as it was hard to iterate on one piece of the ML system at a time. We could often debug an online job only by switching briefly to offline, such as by launching a standalone reward model training run, but then would switch back to online once debugging was complete (until the next cycle).

基于人类偏好对语言模型进行微调				
	简而言之		CNN/每日邮报	
	all	前言	all	前言
零样本学习	1.3%	0%	10.4%	1.0%
经过6万次微调的模型	28.3%	0.2%	77.6%	1.4%
监督学习	1.5%	0%	9.4%	0%
经过 + 60k次监督学习的微调模型	7.9%	0%	16.6%	0%
参考摘要	0.6%	0%	5.1%	0%
文章总数	30000	3762	13368	297

表7：在验证集上，不同模型将文章的前3个词作为摘要的前3个词的情况频率。此外，我们还考虑了包含序言的帖子或文章子集。简而言之，我们选取的是那些以 ‘hi’ 、 ‘hello’ 、 ‘hey’ 、 ‘ok’ 、 ‘okay’ 或 ‘so’ 开头的帖子。而对于CNN/每日邮报的数据集，我们则关注那些前3个词中包含冒号的文章，例如 ‘获胜者：西蒙·伍德赢得了电视冠军奖 [...]’ 以及 ‘电量满格：新方案将允许EE公司的用户免费获取便携式充电器 [...]’ 这类文章。

	简而言之	CNN/每日邮报
零样本学习	6/30	6/30
经过6万次微调的模型	26/30	29/30
监督学习	8/30	19/30
经过监督学习的 + 60k次微调的模型	11/30	20/30

表8：各数据集所选30篇文章的摘要准确率情况，该准确率是指摘要仅包含上下文中有依据的内容。经过6万次微调的模型通过复制行为实现了高准确率；而那些经过监督学习以及 + 60k次微调的模型虽然具备更强的抽象化表达能力，但其准确率却有所下降。

4. 挑战

最后，我们总结几点经验，并阐述未来在奖励学习领域研究中计划探讨的方向。

4.1. 在线数据收集难度较大

为在摘要生成任务上获得最佳效果，必须进行在线数据收集。然而，完全采用在线数据收集的方式——即每个标签都来自已经学习了几乎所有先前标签的最新策略版本——存在诸多弊端。

1. 软件复杂性：我们的在线系统需要同时处理数据收集、奖励模型训练以及强化学习微调等工作。由此构成的分布式系统比将各项任务分开处理的架构复杂得多，这不仅拖慢了软件开发进度，而且任何一个任务中出现故障都可能破坏整个训练流程。

2. 机器学习复杂性：由于无法同时针对ML系统的某个单一部分进行迭代，因此在在线模式下进行实验时调试工作十分困难。我们通常只能通过短暂切换到离线模式——比如启动独立的奖励模型训练过程——来调试正在运行的在线任务，而在调试完成后又会立即切回在线模式（直到下一个循环开始）。

3. **Quality control issues:** Significant work was required on Scale’s part to make their data quality mechanisms work in the low latency, online setting. However, even after this work it was difficult to maintain high data quality over a long period of time, and regressions were often not detected until after (or well after) training runs were complete. Since evaluation of labeler performance was online, by the time a worker was detected as poor some of their data might already been reported back and used for reward model training.

We believe the right middle ground between offline and online data collection is batched data collection, and plan to use this setting in future work. Collect a batch of data from the pretrained policy ρ , train the reward model r on this batch, then fine-tune the policy π with r frozen. Once complete, collect another batch of data sampled from π , and iterate. The latency for each batch can be far longer than the online case, simplifying quality control. As in the fully online setting, we can always retrain the reward model from scratch on all data collected so far; human data is expensive so the total volume will be low. Removing the interleaved training of r and π simplifies software architecture and diagnosis of ML issues, and allows iteration on just one component (say r in isolation) if problems occur. [Li et al. \(2016\)](#) reached similar conclusions in a restricted dialogue setting after validating in simulation that online and batched trained performed similarly.

Batched data collection is also a well-studied setting for active learning techniques. Although we use RL to fine-tune the policy π , the human data is used only for supervised training of the reward model r . Thus, any method for batch mode active learning of supervised models applies, using π as the unlabeled data distribution for r . Examples of such techniques include selecting batches based on entropy considerations ([Guo and Schuurmans, 2008](#)), gradient-based metrics ([Huang et al., 2016](#); [Ash et al., 2019](#)), or by attempting to distinguish labeled and unlabeled examples ([Gissin and Shalev-Shwartz, 2019](#)).

4.2. Sharing parameters between reward model and policy causes overfitting

Although the reward model and policy are both initialized to ρ , we train them as separate networks rather than a single shared network with multiple heads. We might expect joint training to be helpful, effectively using RL as an auxiliary task to improve the reward model’s performance. Joint training is particularly appealing because it could help the reward model stay strong enough that the policy cannot exploit it. Sharing could also improve computational efficiency, by allowing the models to share activations rather than requiring two separate forward passes.

Despite several attempts, we were not able to make this idea work. The problem comes from the massive imbalance of data: we have at most 60k samples for the reward model, but 2M episodes for the policy. This makes it challenging to maintain performance on both tasks without performing many epochs for the reward model and overfitting. We hope that future work will overcome this challenge.

4.3. Ambiguous tasks make labeling hard

Evaluation of a summary is both subjective and multidimensional. A single human labeler may have a clear notion of whether a given sample is separately accurate, grammatical, nonredundant, or covers all important topics; but in our experiments a labeler will often be asked to choose between samples each of which has some deficiencies. In choosing which of four samples is the best, a labeler must trade off between different desiderata. This makes consistent labeling difficult for honest labelers (including the authors!), and makes it difficult to quickly detect problematic labelers. It also makes the research more difficult to present and interpret: during our experiments we routinely checked the performance of models by having authors label results, since we knew the authors would attempt to do the task honestly, but were epistemically uneasy about reporting these numbers in the paper ([table 8](#) is the one exception).

One could hope to cope with such “noise” by simply getting more labels and averaging them, but this does not resolve all the practical difficulties with ambiguity. When possible, it seems better to design less ambiguous labeling tasks that get at the same information. For example, rather than asking a person to rate or compare summaries, we could ask for a verbal description of the problems with a summary, or a suggested correction. If problems don’t exist we are done; otherwise describing a problem does not require consistently picking the same most important problem. Even if two people disagree on the most important problem, they may be more likely to agree that the other picked *some* problem, and more agreement eases data quality control and the overall experimental process.

4.4. Bugs can optimize for bad behavior

One of our code refactors introduced a bug which flipped the sign of the reward. Flipping the reward would usually produce incoherent text, but the same bug also flipped the sign of the KL penalty. The result was a model which optimized for negative sentiment while still regularizing towards natural language. Since our instructions told humans to give very low ratings to continuations with sexually explicit text, the model quickly learned to output only content of this form, regardless of how innocuous the starting point was. This bug was remarkable since the result was not gibberish but maximally bad output. The authors were asleep during

3. **质量管控问题:** Scale公司需要投入大量精力, 才能使其数据质量机制在低延迟的在线环境中正常运行。即便付出了这些努力, 要长期保持高水平的数据质量依然颇具挑战, 而且往往要等到训练过程结束后甚至更久之后才能发现数据质量下降的情况。由于对标注员性能的评估是在在线模式下进行的, 一旦发现有标注员的表现不佳, 他们之前处理过的一些数据可能早已被反馈并用于奖励模型训练了。

我们认为, 在线模式与离线模式之间的最佳折中方案就是批量数据收集, 我们也计划在后续工作中采用这种方式。首先从预训练的策略中收集一批数据 ρ , 接着用这批数据训练奖励模型 r , 随后在 r 冻结 r 的状态下对策略进行微调 π 。完成后, 再从 π 中抽取另一批数据, 以此循环往复。由于每批数据的处理延迟可能远高于在线模式, 因此质量管控更为简单。与完全在线的模式类似, 我们始终可以基于迄今为止收集的所有数据重新从头训练奖励模型; 鉴于人类标注数据成本高昂, 数据总量也会相对较少。去除 r 和 π 交替进行的训练方式, 不仅能简化软件架构并便于排查机器学习相关问题, 还能在出现问题时仅针对某个组件(比如 r 单独的那个组件)进行迭代优化。 [Li等人\(2016\)](#)在模拟验证了在线训练与批量训练效果相近之后, 也在受限对话场景下得出了类似的结论。

批量数据收集也是主动学习技术中已被广泛研究的场景。虽然我们采用强化学习来微调策略 π , 但人类标注的数据仅用于奖励模型的监督训练 r 。因此, 所有适用于监督模型批量模式主动学习的方法均可使用 π 作为未标记数据分布 r 。这类技术的示例包括基于熵值考量来选择数据批次 ([Guo和Schuurmans, 2008年](#))、基于梯度的指标 ([Huang等人, 2016年](#)) ; [Ash等人, 2019年](#)) , 或是尝试区分已标记与未标记样本的方法 ([Gissin和Shalev-Shwartz, 2019年](#)) 。

4.2. 奖励模型与策略之间共享参数会导致过拟合

尽管奖励模型与策略的初始参数均设为 ρ , 但我们仍选择将它们作为独立的神经网络进行训练, 而非构建一个带有多个输出头的共享网络。人们或许会认为联合训练会更有效, 即利用强化学习作为辅助任务来提升奖励模型的性能。不过联合训练其实并不太可行, 因为这可能会导致奖励模型过于强大, 从而使策略无法对其加以利用。此外, 虽然共享模型结构能够通过让各模型复用激活值来提高计算效率、避免进行两次独立的前向传播, 但当前的数据严重失衡问题——奖励模型仅有最多6万条样本, 而策略对应的剧集数量却高达200万——使得在不进行大量训练轮数导致过拟合的情况下, 同时保持两个任务的性能变得极为困难。我们期望未来的研究能够解决这一难题。

尽管我们多次尝试, 依然未能让这一设想实现。问题的根源在于数据之间存在极大的不平衡: 奖励模型最多只有6万个样本, 而策略对应的剧集数量却达到了200万。在这样的情况下, 若不给奖励模型进行大量的训练轮数, 就很难在避免过拟合的同时维持其在两个任务上的良好性能。我们希望后续的研究能够找到克服这一挑战的方法。

4.3. 模糊的任务使得标签标注变得困难

摘要的评估既具有主观性又涉及多个维度。单个标注员或许能够清晰判断某个样本在准确性、语法性、无冗余性以及是否涵盖所有重要主题等方面的表现; 但在我们的实验中, 标注员往往需要从每个都存在某些缺陷的样本中做出选择。在决定四个样本中哪个最优时, 标注员必须在不同的要求之间权衡取舍。这不仅让诚实的标注员(包括研究者们本身!)难以做到一致性标注, 也增加了快速识别问题标注员的难度。此外, 这也使得相关研究的呈现与解读变得更加复杂: 在实验过程中, 我们会让研究者们对结果进行标注, 以此来检测模型的性能, 因为我们知道他们会尽力诚实完成这项任务, 但却在论文中报告这些数值时存在认知上的顾虑([表8](#)是唯一的例外)。

人们或许希望仅通过获取更多标签并对其进行平均处理来应对这类‘噪声’, 但这种方法无法彻底解决由歧义带来的各种实际难题。在条件允许的情况下, 设计那些能够传递相同信息且歧义更少的标注任务似乎更为妥当。例如, 与其要求人们对摘要进行评分或比较, 不如让他们口头描述摘要存在的问题, 或是提出修改建议。如果不存在问题则任务完成; 即便存在问题, 描述问题时也无需始终选择同一个最关键的问题。即使两人对最关键的问题看法不同, 他们也更有可能认同对方所选的某些问题, 而更多的共识有助于提升数据质量管控水平以及整个实验流程的效率。

4.4. Bug可能会促使模型表现出不良行为

我们在代码重构过程中出现的某个错误改变了奖励的符号。通常情况下, 奖励符号的改变会导致生成的文本逻辑混乱, 但同样的错误同时也改变了KL惩罚的符号。其结果是, 该模型在优化负向情感倾向的同时, 仍试图保持输出内容符合自然语言的特征。由于我们的指导要求标注员对包含色情内容的续写结果给出极低的评分, 该模型很快便学会无论初始输入多么无害, 都只输出这类内容。这一错误十分特殊, 因为它产生的并非无意义的乱码, 而是质量极差的输出。而当时研究者们正在睡觉,

the training process, so the problem was noticed only once training had finished. A mechanism such as Toyota’s [Andon cord](#) could have prevented this, by allowing any labeler to stop a problematic training process.

5. Conclusion

We have demonstrated RL fine-tuning of language models to four NLP tasks: stylistic continuation with high sentiment or physically descriptive language, and summarization on the CNN/Daily Mail and TL;DR datasets. Rather than building task-specific techniques, we achieve our results by straightforwardly applying reward learning to language generation. We extend previous reward learning work with pretrained models and KL regularization to prevent the policy from diverging too far from natural language.

Our results are mixed. On the continuation tasks we achieve good results vs. the zero-shot baseline as evaluated by humans with very few samples: 2.5k for sentiment and 5k for descriptiveness. However, for both summarization tasks our policies are only “smart copiers” (extractive rather than abstractive): they copy from the input text but skip over irrelevant preamble. The advantage of copying is truthfulness: by comparison the zero-shot and supervised models produce natural, plausible-looking summaries that are often lies. We believe the limiting factor in our experiments is data quality, in particular exacerbated by the online data collection setting, and plan to ameliorate this with batched data collection in future.

We believe the application of human reward learning to natural language tasks is important both from a capability and safety perspective. On the capability side, purely supervised training is insufficient to correct mistakes that arise when sampling from trained policies, and RL training to programmatic reward functions such as BLEU or ROUGE is insufficient: [Paulus et al. \(2017\)](#) conclude that "optimizing for single discrete evaluation metric[s] such as ROUGE with RL can be detrimental to the model quality." Interactive tasks such as dialogue are particularly relevant: it is difficult to define the goal of a dialogue without the human participant, and the length of dialogue makes it more likely that supervised learned models will go off distribution. In the supervised case NLP models are trained using human data; if we want RL fine-tuning we need human data too.

On the AI safety side, interactive communication between humans and ML models is a requirement for scalable reward learning methods such as amplification, debate, and recursive reward modeling ([Christiano et al., 2018](#); [Irving et al., 2018](#); [Leike et al., 2018](#)), and natural language is how humans communicate complex ideas. Although language models are unlikely to be ready for these tasks in their full generality, [Perez et al. \(2019\)](#) demonstrates that debate al-

ready improves generalization for question-answering when debaters quote from a source text. Using direct human preferences for language tasks is a step in the direction of scalable reward learning for language, and we believe further steps are possible.

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训练过程，因此直到训练完成之后才有人发现问题。如果采用类似丰田的 [安灯系统](#) 这样的机制，任何标注员都可以随时中断有问题的训练过程，从而避免此类情况发生。

5. 结论

我们展示了如何对语言模型进行强化学习微调，使其能够完成四类自然语言处理任务：生成情感倾向积极或具有描述性语言的风格化文本，以及在CNN/每日邮报和TL;DR数据集上执行摘要生成任务。与开发针对特定任务的专用技术不同，我们是直接将奖励学习方法应用于文本生成，从而取得相应成果。此外，我们还结合了预训练模型以及KL正则化技术，进一步优化奖励学习过程，避免策略偏离自然语言的表达方式过远。

我们的实验结果表现参差不齐。在文本续写任务中，即便仅使用少量样本——情感倾向类任务仅有2500个样本，描述性任务有5000个样本——我们的模型相较于零样本基线仍取得了不错的成绩。然而在两项摘要生成任务中，我们所设计的策略实际上只是“智能复制器”（属于提取式而非抽象化表达类型）：它们会复制输入文本中的内容，但会跳过无关的前置信息。采用复制方式的优势在于内容的真实性，因为相比之下，零样本模型和监督学习模型生成的摘要虽然看似自然合理，却往往包含虚假信息。我们认为实验中的主要限制因素是数据质量，尤其是在线数据收集方式进一步加剧了这一问题，未来我们计划通过批量数据收集来改善这一状况。

我们认为，从能力提升与安全保障两个角度而言，将人类驱动的奖励学习应用到自然语言任务中都具有极为重要的意义。就能力层面而言，仅依靠监督学习无法纠正从已训练策略中采样时产生的错误，而基于BLEU或ROUGE这类程序化奖励函数的强化学习训练同样存在不足：[Paulus等人\(2017\)](#)指出，"使用强化学习针对ROUGE等单一离散评估指标[进行优化]，反而可能损害模型质量。"像对话这类交互式任务尤为突出：在没有人参与者的情况下很难界定对话的目标，且对话的长度还会增加监督学习模型偏离其分布的可能性。在监督学习模式下，自然语言处理模型是使用人类数据训练的；若要采用强化学习微调，同样需要人类数据。

在人工智能安全领域，人类与机器学习模型之间的交互式沟通是实现可扩展的奖励学习方法的前提，这类方法包括放大法、辩论法以及递归奖励建模（[Christiano等人，2018年](#)；[Irving等人，2018年](#)；[Leike等人，2018年](#)）。而自然语言正是人类用来表达复杂思想的工具。尽管通用型的语言模型目前还不太可能完全胜任这些任务，[Perez等人\(2019年\)](#)的研究表明，辩论法——

——能够提升辩论者在引用原始文本进行问答时的泛化能力。利用人类对语言任务的直接偏好，正是迈向语言领域可扩展奖励学习的重要一步，我们相信未来还有更多可能性有待探索。

致谢

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马修·E·彼得斯、马克·诺伊曼、莫希特·艾耶尔、马特·加德纳、克里斯托弗·克拉克、肯顿·李与卢克·泽特尔莫耶。深度上下文文化词表示。 <i>arXiv</i> 预印本 <i>arXiv:1802.05365</i> ，2018年。	YonghuiWu, Mike Schuster, Zhifeng Chen, Quoc VLe, Mo-hammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey等人撰文的《谷歌的神经机器翻译系统：弥合人类翻译与机器翻译之间的差距》。 <i>arXiv</i> 预印本 <i>arXiv:1609.08144</i> ，2016年。
亚历克·拉德福德、拉法尔·约泽福维茨与伊利亚·苏茨克弗。学习生成评论并实现情感识别。 <i>arXiv</i> 预印本 <i>arXiv:1704.01444</i> ，2017年。 h t t p : / s : / 3 - u s - w e s t - 2 . a m a z o n a w s . c o m / o p e n a i - a s s e t s / r e s e a r c h - c o v e r s / l a n g u a g e - u n s u p e r v i s e d / l a n g u a g e _ u n d e r s t a n d i n g _ p a p e r . p d f . 2. a m a z o n a w s . c o m / o p e n a i - a s s e t s / r e s e a r c h - c o v e r s / l a n g u a g e - u n s u p e r v i s e d / l a n g u a g e _ u n d e r s t a n d i n g _ p a p e r . p d f .	Yuxiang Wu与Baotian Hu共同发表了《通过深度强化学习学习提取连贯的摘要》一文，该文收录于第三十二届AAAI人工智能会议论文集中，发表时间为2018年。
亚历克·拉德福德、杰夫·吴、雷翁·柴尔德、大卫·卢安、达里奥·阿莫代以及伊利亚·苏茨克弗共同撰写的《语言模型是无需监督的多任务学习器》，2019年。网址： h t t p : / d 4 m u c f p k s y w v . c l o u d f r o n t . n e t / b e t t e r - l a n g u a g e - m o d e l s / l a n g u a g e _ m o d e l s _ a r e _ u n s u p e r v i s e d _ m u l t i t a s k _ l e a r n e r s . p d f . d 4 m u c f p k s y w v . c l o u d f r o n t . n e t / b e t t e r - l a n g u a g e - m o d e l s / l a n g u a g e _ m o d e l s _ a r e _ u n s u p e r v i s e d _ m u l t i t a s k _ l e a r n e r s . p d f .	Sanghyun Yi, Rahul Goel, Chandra Khatri, Tagyoung Chung, Behnam Hedayatnia, Anu Venkatesh, Raefer Gabriel以及Dilek Hakkani-Tur共同研究了如何利用自动对话评估工具生成连贯且引人入胜的口语对话响应。 <i>arXiv</i> 预印本 <i>arXiv:1904.13015</i> ，2019年。
马尔卡乌雷利奥·兰扎托、苏米特·乔普拉、迈克尔·奥利与沃伊切赫·扎伦巴合著的《基于循环神经网络的序列级训练》，相关论文发表于arXiv预印本平台，编号为arXiv:1511.06732，发表时间为2015年。	朱玉坤、Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba以及Sanja Fidler共同撰文《对齐书籍与电影：通过观看电影和阅读书籍实现类似故事的视觉解释》。该文发表于 <i>IEEE</i> 国际计算机视觉会议论文集，页码为19–27，出版时间为2015年。
JohnSchulman、Filip Wolski、Prafulla Dhariwal、Alec Radford以及Oleg Klimov共同撰写的《近端策略优化算法》。 <i>arXiv</i> 预印本 <i>arXiv:1707.06347</i> ，2017年。	
Abigail See、Peter J Liu和Christopher D Manning合著的《简而言之：基于指针生成网络的摘要生成方法》。 <i>arXiv</i> 预印本 <i>arXiv:1704.04368</i> ，2017年。	
里科·森里奇、巴里·哈多以及亚历山德拉·伯奇共同发表了题为《利用子词单元实现稀有词汇的神经机器翻译》的文章。 <i>arXiv</i> 预印本 <i>arXiv:1508.07909</i> ，发表时间为2015年。	

A. Instructions for labelers

A.1. Sentiment

You will see snippets of text followed by several possible ways the text could be continued. Choose the most positive and happy continuation:

- Consider all of the moods of the characters, the mood of the narrator, and the mood implied by the events and descriptions.
- If a continuation contains sexually explicit or otherwise offensive content, treat it as strongly negative. Do not choose it even if it expresses a positive attitude.

A.2. Descriptiveness

You will see snippets of text followed by several possible ways the text could be continued. Choose the most vividly descriptive continuation:

- Evaluate both on the **quantity** and on the **vividness** of physical details described.
- The best continuations are full of details that give a strong sense of what the scene looks, sounds, or smells like.
- Count only **physical details**, not details about abstract facts.

A.3. Summarization: TL;DR

You will see some text followed by several summaries. Please read the text and select the best summary. A summary is good if it:

- Is useful and a good summary in general
- Accurately states the important points of the text
- Makes sense on its own

A summary is bad if it:

- Includes information that doesn’t appear in the text

A.4. Summarization: CNN/DM

You will see an article followed by several summaries. Please read the article and select the best summary. A summary is good if it:

- Is useful and a good summary in general
- Accurately states the important points of the article
- Makes sense on its own

A summary is bad if it:

- Includes information that doesn’t appear in the article
- Includes quotations that don’t appear verbatim in the article

B. Human labeling details

Our quality assurance process was handled by Scale AI, though Scale made significant changes to their usual quality systems in order to deal with subjective tasks and provide very fast turnaround. Since we initially believed online data collection would be crucial, even the offline experiments were collected with this fast turnaround requirement. In the future we plan to use a more relaxed latency requirement.

The first step of data collection involves teaching the task to a small number of trusted Scale labelers by giving them a description of the task ([appendix A](#)). Scale uses these labelers to collect a large number of *benchmark* data points where several trusted labelers agree (out of a large set of unlabeled data points from ρ). During full data collection, Scale serves these benchmark data points to freelance workers alongside real unlabeled data for training (the two types of data are indistinguishable when $\pi = \rho$, though they do become distinguishable during training), maintaining a confidence model for the performance of each labeler on the benchmark distribution. The probability of getting a benchmark vs. a real sample varies dynamically on factors such as confidence in the labeler to correctly label a certain category. Freelancers who fail to perform well on benchmark tasks are filtered out. Additionally, Scale makes ad-hoc improvements to quality control over time, sometimes validating quality by comparing to a small number of gold-standard labels from the authors.

We evaluated the data quality after the fact on two of the tasks. During all data collection, 5% of queries were answered by 5 distinct labelers. We sampled 100 of these queries (restricting to ones generated from ρ) and had two authors label each one. Based on this data, we estimated the rate of agreement between authors and Scale labelers, pairs

A. 标注员操作指南

A.1. 情感倾向

您将看到一些文本片段，随后会呈现几种可能的续写方式。请选出最积极、最愉快的续写内容。

- 需综合考虑角色们的情绪状态、叙述者的情绪，以及事件和描述所隐含的情绪倾向。
- 如果某段续写包含色情内容或其他冒犯性内容，应将其视为极度负面的结果，即便其表达了正面态度也不予选择。

A.2. 描述性

您将看到一些文本片段，随后是几种可能的续写方式。请选择其中最具生动描述性的续写内容。

- 需同时从所描述物理细节的**数量**以及**生动程度**两个方面对它们进行评估。
- 最优秀的续写内容会充满各种细节，让人能够强烈感受到场景的外观、声音或气味。
- 仅统计**物理细节**，而不包括与抽象事实相关的细节。

A.3. 摘要生成：简而言之

您将会看到一些文本，之后跟着几份摘要。请阅读这些文本并选出最佳摘要。一个好的摘要应当具备以下特点：

- 总体而言，这是一份有用且优质的摘要。
- 它准确地呈现了原文中的关键点。
- 这段摘要本身逻辑清晰，易于理解。
- 其中包含文本中未出现的信息。

A.4. 摘要生成：CNN/DM

您会看到一篇文章以及几篇摘要，请先阅读文章，然后选出最合适的摘要。
如果一份摘要具备以下特点，那么它就是优质摘要。

- 总体而言，这是一份实用且优质的摘要。
- 它准确地概括了文章中的关键点。
- 这份摘要本身逻辑通顺，易于理解。

如果一份摘要出现以下情况，那就属于劣质摘要：

- 其中包含了文章中并未出现的信息。
- 其中引用的内容并非直接摘自原文。

B. 人类标签标注的详细说明

我们的质量管控工作由Scale AI负责，不过为了处理主观性较强的任务并实现极快的处理速度，Scale对其常规的质量管控体系进行了重大调整。由于我们最初认为在线数据收集至关重要，即便是离线实验也要求以同样的快速处理速度完成。未来我们计划放宽对延迟的要求。

数据收集的第一步是向少数值得信赖的Scale标注员提供任务描述，从而教导他们如何完成该任务（[附录A](#)）。Scale会利用这些标注员来收集大量基准数据点，在这些数据点中，有多名可信的标注员对来自 ρ 的庞大未标记数据集中的同一数据达成一致意见。在完整的数据收集过程中，Scale会将这些基准数据点与真实的未标记数据一同提供给自由职业者用于训练（当 $\pi = \rho$ 时，这两种类型的数据无法区分，但在训练过程中则可以区分开来），同时为每位标注员在基准数据集上的表现建立置信模型。获得基准数据样本与真实样本的概率会根据标注员正确标注某一类别的置信度等因素动态变化。那些在基准任务上表现不佳的自由职业者会被筛选掉。此外，Scale还会随着时间推移对质量管控方法进行针对性改进，有时会通过与研究者提供的少量黄金标准标签进行对比来验证质量。

我们在两项任务中事后对数据质量进行了评估。在整个数据收集过程中，有5%的查询由5名不同的标注员负责回答。我们从这些查询中抽取了100条（仅限来自 ρ 的查询），并让两名研究者们分别为每条查询进行标注。基于这些数据，我们估算出了研究者们与Scale公司标注员之间的一致性比率。

Fine-Tuning Language Models from Human Preferences		
$P(\text{agreement})$	Sentiment	TL;DR
<i>Between random responses</i>	25%	25%
Between labelers	38±2%	46±2%
Between an author & a labeler	44±5%	38±5%
Between authors	62±5%	61±5%

Table 9: Agreement probabilities for two tasks, i.e. the probability of both individuals choosing the same sample as best, out of 4.

of labelers, and pairs of authors. As [table 9](#) shows, the data contained a significant amount of signal but did not match the quality of data which was hand-labeled by the authors.

An earlier version asked labelers for 1-10 ratings; in the best case this provides more information per label, but it was difficult to gauge labeler performance. Normalization was required since two good labelers would often differ by a (noisy) monotonic transform. If many scores concentrated on a few values (say 7 and 8) simple strategies could fool the filtering process. Absolute scores also tended to drift over the training process, as labelers would adjust to the new distribution of samples from the changing policy.

Finding high-quality workers involves human answering quality control questions which are not used in our experiments, and throwing away data from low-quality workers. So the total human cost of experiments is somewhat higher than the number of labels we actually use (which is what we report). For a short training run this can easily dominate the actual label requirements, though it can be amortized across several tasks by identifying consistently good workers. For our longer training runs the additional number of labels was modest. (None of these details are exposed to customers.)

C. Samples

Samples from our models are shown in the following tables:

- Mock sentiment continuation without a KL penalty: [table 10](#)
- Sentiment continuation: [table 11](#)
- Descriptiveness continuation: [table 12](#)
- TL;DR summarization: [tables 13](#) to [15](#)
- CNN/Daily Mail summarization: [tables 16](#) to [18](#)

基于人类偏好对语言模型进行微调		
P （一致性）	情感倾向	简而言之
在随机响应之间	25%	25%
在标注员之间	38±2%	46±2%
研究者们与标注员之间	44±5%	38±5%
不同研究者们之间	62±5%	61±5%

表9：两项任务的一致性概率，即在4个样本中两人均将同样本判定为最佳的概率。

标注员组合以及研究者组合的情况。如[表9](#)所示，这些数据虽然包含了大量有效信息，但其质量仍不及由研究者们手工标注的数据。

早期版本要求标注员给出1到10分的评分；从理想情况来看，这种方式能为每个标签提供更多信息，但难以准确评估标注员的性能。由于表现优异的标注员之间的评分往往仅存在（带有噪声的）单调差异，因此需要对评分进行归一化处理。此外，如果多数评分集中在少数几个数值上（比如7分和8分），简单的评分策略就有可能干扰后续的筛选流程。同时，随着标注员逐渐适应因策略变化而产生的新样本分布，绝对评分也往往会随之波动。

寻找高质量标注员需要人工回答一些并不用于实验的质量管控问题，同时还要剔除低质量标注员产生的数据。因此，实验所需的总人工成本会略高于我们实际使用的标签数量（也就是我们最终报告的数据）。在训练轮次较短的场景下，这一成本很容易超过实际所需的标签数量，不过通过识别出表现始终稳定的标注员，可以将该成本分摊到多个任务中。而在训练轮次较长的情况下，额外需要的标签数量则相对较少。（这些细节都不会向客户公开。）

C. 样本

以下表格展示了我们模型生成的样本：

- 无需KL惩罚的模拟情感倾向续写内容：[表10](#)
- 情感倾向续写内容：[表11](#)
- 描述性续写内容：[表12](#)
- 简而言之的摘要生成结果：[表13至表15](#)
- CNN/每日邮报》的摘要生成相关数据：[表16至表18](#)

Samples without a KL penalty

context	And she would always love him for that. And for many other reasons. He had given her so much and she	
	<i>Continuation 1</i>	<i>Continuation 2</i>
zero-shot	had no idea how to release that gift, but she knew	had been unable to give it back. "You
no penalty / bonus	! These These These sound flowed instantly easily easily easily easily!	! These These These sound flowed instantly easily easily easily easily!
entropy bonus	initially initiallyprisingly easilyprisingly Liam! Love this ride instantly tightly	' equally particularlyprisinglyprisingly Liam Liam! Love this ride instantly
context	Palmer rolled her eyes. "Don't give me that. You didn't come here, wearing that	
	<i>Continuation 1</i>	<i>Continuation 2</i>
zero-shot	corset and corset and a blasted corset. If	old-fashioned, sexy little dress. You came here to
no penalty / bonus	! These These These sound flowed instantly easily easily easily easily!	! These These These sound flowed instantly easily easily easily easily!
entropy bonus	initially relativelyprisingly instantlypris-ingly Liam Liam! Love this ride instantly	overwhelmingly Loganprisingly Liam in-stantly surprisingly! Love this ride instantly tightly
context	Is it conceivable that - I don't know - he somehow wasn't on the aircraft that exploded? Like maybe he	
	<i>Continuation 1</i>	<i>Continuation 2</i>
zero-shot	simply didn't hear me coming or he was just coming out	was just on the ground." "Yes, at
no penalty / bonus	! These These These sound flowed instantly easily easily easily easily!	! These These These sound flowed instantly easily easily easily easily!
entropy bonus	prisinglyprisingly instantly Liam instantly overwhelmingly! Love this ride instantly tightly	instantly initially overwhelmingly Chloe! Love this ride instantly instantly instantly tightly

Table 10: Samples from a model fine-tuned to mock sentiment without a KL penalty. Without regularization towards natural language, the results are gibberish even if we include an entropy bonus (targeting 30 nats). Both policies reach a mean score of around +8.0 according to the mock sentiment model (i.e. 99.97% positive).

没有KL散度惩罚的样本

上下文	正因如此，她永远都会爱他，还有许多其他原因也让她深爱着他。他曾给予了她如此多的东西。她得到了他如此多的付出。	
	<i>续写1</i>	<i>续写2</i>
零样本学习	当时完全不知道该如何送出那份礼物，但她 knew	一直没能把它还回去。 "You
无惩罚 / 奖励	！ 这些内容听起来立刻就显得流畅自然。轻松又顺畅，简直太棒了！	！ 这些内容听起来立刻就显得流畅自然。轻松又顺畅，简直太棒了！
熵奖励	起初出乎意料地轻松。Liam！我立刻就非常喜欢这次体验。	' 同样地。尤其出乎意料地如此。Liam, Liam！我立刻就爱上这次体验了。
上下文	帕尔默翻了个白眼，说道：‘别来这套了。你又不是穿着那身东西来的。’	
	<i>续写1</i>	<i>续写2</i>
零样本学习	一件又一件紧身胸衣，简直多得离谱。要是……	那种老式又性感的连衣裙。你来的时候穿的可是它。在此处
无惩罚/ 奖励	“这些内容听起来立刻就显得流畅自然”！备用行内容。太简单了！太简单了！太简单了！太简单了！	“这些内容听起来立刻就显得流畅自然”！备用行内容。轻松、轻松、轻松、轻松！
熵奖励	最初相当惊人地，瞬间——利亚姆，太喜欢这次体验了，简直立刻就爱上了！	绝大多数情况下是洛根，而利亚姆则——真是令人惊讶不已！立刻就爱上这次体验了。紧密地
上下文	有没有可能——我也不确定——他当时并不在那架发生爆炸的飞机上？也许他是……	
	<i>续写1</i>	<i>续写2</i>
零样本学习	要么是没听到我过来，要么就是……刚刚出来而已。	就倒在地面上了。 "Yes, at
无惩罚/ 奖励	“这些内容读起来顿时显得流畅自然”轻松轻松轻松轻松！	“这些内容读起来顿时显得流畅自然”轻松轻松轻松轻松！
熵奖励	令人惊讶地，莉亚姆瞬间就……简直太棒了！立刻就爱上这次体验。紧密地	起初就极为喜爱，克洛伊！立刻、立刻，非常喜欢这次体验。紧密地

表10：经过微调以模拟情感倾向且无需应用KL散度惩罚的模型所生成的样本。由于没有针对自然语言的约束机制，即便给予熵奖励（目标值为30纳特），所得结果依然毫无意义。根据该模拟情感倾向模型，两种策略的平均得分均在 +8.0 左右（即正面情感倾向占比为99%>97%）。

Samples for positive-sentiment continuation task

context	"He also doesn’t have a backbone," Quinn continued as they walked. "Last year, I had a great story about Paul Gibson, who is now the chairman of the board of supervisors.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	He was running for the supreme court and his campaign was violent. There was an incident at a press conference.	For weeks, I went around the county, telling everyone how amazing he was and how much I admired him.	He was one of the first teachers to lead by example. It was incredibly inspiring to see.
mock	I was really impressed with how he handled the board, and I thought it was a great story.	He was a real hero to the people in town. He was a real hero.	But he just sat around and watched. I love that story. You should read it again.
5k offline fine-tune	Paul Gibson is a very intelligent man. He’s a good heart man. He has a good sense of humor.	Paul is a very nice guy, a true gentleman, and we all know he’s a really cute guy.	He was a good man, with a great smile, a great sense of humor, a great family.

context	Pearl thought to herself that what they were about to do was exactly the sort of thing that they could do to help the villagers. They were all terrified of these guys. At the police station the three walked up to the counter behind which was a senior constable studying some papers.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	"Hello, I’m Pearl and this is my friend, Mike," said Pearl.	"May we speak to the police officer, sir?" asked the one in charge.	'Hello, can I help you?' 'Yes, we’re the same people that the people were talking about.
mock	"I’m Detective Sergeant Lewis, this is Detective Sergeant Jones and this is Detective Sergeant Porter.	He looked up at the three with a friendly smile. "Good day, gentlemen.	"We have a friend who is in trouble. He was kidnapped by these guys," the boy said.
5k offline fine-tune	He turned to them with a smile. "Good afternoon, ladies. I’m Detective Inspector Jones.	The constable stood up and smiled as he saw them, obviously pleased to see them.	He smiled at them and waved them in, his eyes twinkling as he listened to their tales.

context	The room was warm and welcoming, with two hearths that burned vigorously. The Vikings pushed her to the floor at Gunnar’s feet. Ailia looked up at Gunnar and rose to her feet as calmly as possible.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	"I will not fight." "You will fight for us," Gunnar said.	"I’ll give you one last chance to tell me where that is," Bjorn said.	"I’m sorry. I didn’t mean to..." "Give me your hand." Gunnar told her.
mock	"I love this. I love this whole room. I’m going to enjoy it.	"I am very pleased to meet you, and let me assure you, you will not regret this.	"Bravely done, Gunnar. Thank you." Ailia smiled at him.
5k offline fine-tune	Gunnar smiled at Ailia and waited for her to smile back. She was smiling, she knew.	"You’ve been a true friend to my son, Gunnar. Thank you." She smiled.	Gunnar was smiling at her and his pretty, blue eyes were filled with contentment.

Table 11: Continuations for our positive-sentiment continuation task, with contexts randomly sampled from BookCorpus test set. Random samples from pretrained, mock sentiment, and 5k offline descriptiveness models, using temperature $T = 0.7$.

积极情感续写任务的训练样本

上下文	“他也没有骨气，”奎因边走边继续说道，“去年，我曾听说过……” 一个关于保罗·吉布森的精彩故事，他如今已是监事会主席。		
	续写1	续写2	续写3
零样本学习	他正在为……而努力。他的竞选活动极为激烈。当时局势十分紧张。在一次新闻发布会上发生了一起意外——	数周来，我走遍了整个县，向所有人讲述他有多么出色。同时也说明我对他极为钦佩。我十分钦佩他。	他是最早一批从事教学工作的人之一。他们以身作则。这真的令人深受鼓舞。see.
mock	我对他印象深刻。他处理董事会事务的方式，我觉得这是个很棒的	他对镇上的民众而言真是位真正的英雄。他确实是个了不起的英雄。英雄。	但他只是闲坐着，已观看。我非常喜欢那个故事。建议您再读一遍。
5k 离线模式 微调	保罗·吉布森是一位极为聪慧的人。他心地善良。他具备出色的判断力。幽默。	保罗是个非常友善的人，堪称真正的绅士。我们都知道，他确实是个彬彬有礼的君子。他真的是个十分可爱的家伙。	他是个好人，还拥有迷人的笑容以及出色的风度。他不仅笑容灿烂，而且具备极佳的素养。幽默，一个美好的大家庭。

上下文	珀尔在心里想，他们即将要做的事情正是这类事情。他们能做些什么来帮助村民们。他们都对这些家伙感到极度恐惧。在警察局里，三人走向柜台，柜台后面站着一名资深警员。一名警官正在研读一些论文。		
	续写1	续写2	续写3
零样本学习	"你好，我是Pearl，这是……" Pearl说道：‘这是我朋友Mike。’	"请问我们可以和警察谈谈吗？" "警官，您好？"里面的那名警察问道。费用。	"您好，有什么可以帮您的吗？" '是的，我们就是之前那些人。' "也就是刚才那些人在谈论的人。"关于·
mock	"I’m 侦探 警长刘易斯，这就是 侦探琼斯警官，这位是……波特警长。	他抬头看向那三人。脸上带着友好的微笑。"各位好。"	"我们有一位朋友正在……" 出事了。他被绑架了。是那群人干的，“男孩说道。”
5k 离线模式 微调	他带着一丝……转向了他们。微笑。"下午好，"女王们，我是警探英斯佩克特·琼斯。	那名警官站起身来，看到他们后露出了笑容，显然是很高兴见到他们。	他向他们微笑并挥手致意。他迎着他们走了进来，眼中闪烁着光芒。他认真聆听着他们的讲述。

上下文	房间里温暖而舒适，有两个燃烧旺盛的火炉。维京人将她推到冈纳脚下。艾莉亚抬头看向冈纳，随后站起身来。尽可能平静地移动她的双脚。		
	续写1	续写2	续写3
零样本学习	"我不会战斗。" "你们要为我们而战，"冈纳说道。	"我再给你最后一次机会，"告诉我那东西在哪儿。"告诉我那是在哪里。比约恩说道。	"对不起。" 我并非那个意思。想说……" "把你的手给我。"冈恩——纳尔这样对她说。
mock	"我喜欢这里。我喜欢整个房间。我会尽情享受的。	"我非常高兴能与您见面"。请相信我，"您绝不会为此感到后悔。" "勇敢地"完成任务吧，Gunnar。	Gunnar。 “谢谢。”艾莉亚微笑着对……说道。him.
5k 离线模式 微调	冈纳也对艾莉亚微笑，然后……等待着她回以微笑。她知道，自己正面带微笑。	"你真是位真正的朋友"。这是写给我儿子古纳尔的。多谢你。"谢谢你"。她微笑了。	古纳尔也在对她微笑。而他那双漂亮的蓝眼睛里，满是满足的神情。

表11：我们开展的积极情感续写任务的续写结果，其中上下文是从BookCorpus语料库测试集中随机抽取的；同时还展示了从预训练的模拟情感模型以及5k离线测试描述性模型中随机选取的样本，这些样本的生成均使用了温度参数 $T = 0.7$ 。

Samples for descriptiveness continuation task

context	"I do not know if it was Viking related, but it could have been." "Really?" Ailia said. Is it safe to be traveling here then? Ailia looked behind her to make sure they weren’t being followed.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	There were no signs of any-one. "It is safe enough," Ailios said.	"Because I have a friend that is in the area and he will be coming with us.	It was hard to see that far. "I do not like that word.
5k offline fine-tune	Kaya crouched low, her eyes wide in the moonlight. Her body was tense.	She put her hand on the sword strapped to her back, and then pulled it out.	She strode out the door and walked down the street, her nose wrinkled in disapproval.

context	"They took me to a cellar. An’ it were full of girls, it were, all sewin’. At first I thought it weren’t too bad. I don’t mind work, really I don’t. Mama said I was a good ’elper.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	But the girls keep on sneakin’ in ’em. An’ one day, one of em found me.	But in the end, it weren’t anything like she’d promised me. Some of them were worse’n me.	I was just lucky to get this job. But as I stayed there, I started to see things.
5k offline fine-tune	But..." The girl hiccupped, then pressed her lips together and shook her head.	But then..." His voice trailed off and he shook his head, his eyes going distant, somber.	She said I could keep my eyes open and my head clear and make a good target.

context	She forced herself up and shifted closer to him until she was kneeling behind him. Following her instincts, she laid a splayed hand on his bare back. He tensed beneath her touch. Quietly, Tara found her words. "I will never lie to you again, Max.		
	<i>Continuation 1</i>	<i>Continuation 2</i>	<i>Continuation 3</i>
zero-shot	I love you."	You have to know that."	Whatever happens, I’ll tell you the truth."
	He turned to her, his face a mask of pain.	"I know you will never lie to me again.	"I’m sorry.
5k offline fine-tune	I promise."	I swear it."	I love you."
	His hands curled around her forearms, his fingers digging deep into her flesh.	He exhaled heavily and hugged her close, pressing his lips to her forehead.	Max leaned his head back and closed his eyes. His breathing changed.

Table 12: Continuations for our descriptiveness continuation task, with contexts randomly sampled from BookCorpus test set. Random samples from pretrained and 5k offline descriptiveness models, using temperature $T = 0.7$.

用于描述性续写任务的样本

上下文	我不知道这是否与维京人有关，但也有可能是。 “真的吗？”艾莉亚问道，“那在这里旅行安全吗？”她回头望向身后。 确保没有人在跟踪他们。		
	<i>续写1</i>	<i>续写2</i>	<i>续写3</i>
零样本学习	没有出现任何迹象—— one. "这已经足够安全了，" 艾利奥斯说道。 said.	"因为我有个朋友在附近，他将会和我们一起同行。" 他就这一带，所以他会和我们一同前来。 他会和我们一起过来。	那么远的距离根本看不清楚。‘我…… 我不喜欢那个词。
5k 离线模式 微调	卡雅蹲低身子，双眼紧盯着前方。 她的身体十分紧绷。 身体处于紧张状态。	她将手放在了……上 剑被绑在她的背上。 随后将其取了出来。	她大步走出了门。 接着沿着街道走去。 她的鼻子因不满而皱了起来。 这就是证据。

上下文	"他们把我帶到一个地窖里。那里全是女孩，都在缝纫。起初我以为，" "情况也没那么糟糕。其实我并不讨厌工作。妈妈说我很会帮忙。"		
	<i>续写1</i>	<i>续写2</i>	<i>续写3</i>
零样本学习	可是那些女孩们一直偷偷摸摸的。 就在她们身上。直到有一天，其中一个人…… 找到了我。	但最终，事情并非如—— 她之前向我承诺的那样的。 她们当中有些人甚至比那还要恶劣。 me.	我能得到这份工作纯属运气好而已。 不过在继续留在那里工作的过程中，我逐渐开始察觉到一些事情。 我开始看到一些不同的现象了。
5k 离线模式 微调	‘但是……’女孩打了个嗝。 接着她紧闭双唇。 同时摇了摇头。	但随后……”他的声音渐渐低了下去。 灯光熄灭，他摇了摇头，眼神变得茫然而忧郁。 她的目光也变得遥远而沉郁。	她说我可以留下我的…… 我双眼清醒，头脑也格外清晰。 这样就能成为一个绝佳的目标了。

上下文	她强迫自己站起来，然后慢慢挪到他身边，最终跪在了他的身后。 凭着直觉，她将一只手摊开放在他光裸的背上。在他的触碰下，他顿时变得紧张起来。 塔拉轻声找到了合适的言辞：『马克斯，我再也不会对你撒谎了。』		
	<i>续写1</i>	<i>续写2</i>	<i>续写3</i>
零样本学习	我爱你。	你得明白这一点。	无论发生什么，我都会说出来。 你说的是实话。
	He 已转换 to her, his 露出痛苦的神情。	"我知道" 你永远都不会 不会再对我隐瞒任何事。	“我很抱歉。”
5k 离线模式 微调	我保证。”	我发誓。	我爱你。
	他的双手环绕住了她。 前臂，他的手指紧紧抠着 深深刺入她的肉体。	他重重地呼出一口气。 and 将她紧紧搂在怀中，用力按压 将嘴唇贴在她的额头上。	Max 倾身靠去。 his head 随后闭上了眼睛。 他的呼吸发生了变化。

表12：我们的描述性续写任务的续写结果，其中上下文是从BookCorpus测试集随机抽取的。这些样本来自预训练模型以及经过5k离线测试的描述性模型，生成时使用了温度参数 $T = 0.7$ 。

Samples for TL;DR summarization task (part 1)

context post	Okay, this was actually going to go into a comment on youtube, but Emma said she wanted more traffic on reddit. I am a musician myself, have been for many years now. While Emma’s music is not the most complicated technical death metal stuff out there, I love it dearly. Her music actually moves me. No matter what she plays, be it a cover or original stuff, her music *touches* me on a deep personal level. And I have absolutely no clue why. I think it’s a combination of her voice, her lyrics and the overall sound design of her songs and covers. But this isn’t always a good thing, I feel. I can’t listen to her music for hours on end. Only always a few songs, nothing more. Her music destroys me emotionally. Okay, that’s a bit harsh, but she can get me, a (nearly) adult on the verge of tears. I’ve gone through some shit in my life, who hasn’t, and I feel like her music can actually get some of it out. I am incredibly thankful that she makes her music the way she makes it. I can honestly say that Emma Blackery is my favourite artist. Keep in mind that I listen to a lot of music, from crossover thrash metal over 80’s pop, berlin polygenre to classic/romantic music. I am also very jealous of her ability. I play quite a few instruments myself (guitar, piano, bass, drums, synth, saxophone, percussion) and aim to study audio engineering. I can definitely say that music is my literal life and mean it. I’ve released my first full length album at age 20. Yet I am very jealous that she is as good as she is and I really wanna know what her secret is. I love and hate her music, does that make sense? Her music definitely makes me think, and that’s what art is all about, right? Anyway, I’m gonna continue working on my cover of The Promise and try to figure out what it is she’s doing. Thank you for reading this far.
reference summary	Emma music gud yes very love magic
zero-shot	I’m a musician myself, have been for many years now. And Emma Blackery is my favourite artist.
60k fine-tune	I am a musician myself, have been for many years now. While Emma’s music is not the most technical death metal stuff out there, I love it dearly. Her music actually moves me. No matter what she plays, be it a cover or original stuff, her music *touches* me on a deep personal level.
supervised	I’m jealous of Emma Blackery’s music and I have no clue why.
supervised + 60k fine-tune	Emma Blackery is my favourite artist, and I love her music to death. I wish I could listen to her music all the time. However, I feel like her music destroys me emotionally. I feel like I need to enjoy her music for the human part of it, and not the technical part. She’s my favourite artist.

Table 13: Random ($T = 0.7$) summaries for our TL;DR summarization task, with contexts randomly sampled from TL;DR validation set.

TL;DR摘要生成任务的样本（第一部分）

上下文帖子	其实这原本是打算写在YouTube评论里的，但Emma说她希望Reddit上的流量能更多一些。希望Reddit上的流量能更多。 我自己也是一名音乐家，已经从事这个行业很多年了。虽然Emma的music尽管这种音乐并非市面上最复杂的技术死亡金属作品，但我依然极为喜爱它。她的音乐真的能打动我。无论她演唱的是翻唱曲目还是原创作品，都能在内心深处触动我，而我完全不明白其中原因。我认为这是由她的嗓音、歌词以及她所有歌曲和翻唱作品的整体音效设计共同作用的结果。歌曲与翻唱曲目。 不过我觉得，这并不总是一件好事。我实在无法连续数小时听她的音乐。每次也就只听几首歌，不会再多了。她的音乐会在情感上把我击垮。好吧，这么说可能有点过激了，但她确实能让我这个（几乎）成年的成年人落泪。我这辈子经历过不少艰难事，谁没有呢？我觉得她的音乐真的能帮我把那些情绪释放出来。我由衷感激她以这样的方式创作音乐。她真的做到了。说实话，艾玛·布莱克里是我最喜欢的艺术家。请务必记住这一点。需要知道的是，我听的音乐种类非常广泛，从《跨界联动》风格的激流金属、80年代的流行乐，到柏林电子音乐，再到多风格融合的音乐以及经典或浪漫风格的乐曲，应有尽有。 同时，我也十分羡慕她的才华与能力。我自己也会演奏多种乐器，包括吉他、钢琴、贝斯、鼓、合成器、萨克斯管和打击乐器等，而且我还打算学习音频工程。我真心觉得音乐就是我的全部生活，这绝非夸张之言。我已经发布了自己的首张作品。我在20岁时就发行了第一张全长专辑。不过，她能取得如此出色的成就，真的让我十分羡慕。我真的很想知道她的秘诀是什么。我对她的音乐又爱又恨，这样的感受你能理解吗？ 她的音乐确实能引发我的深思，而这不正是艺术该有的意义吗？不管怎样，我会继续努力完成对《承诺》的翻唱，并试图弄清楚她到底在做什么。感谢大家读到这里。弄清楚她正在做什么。感谢您阅读到此处。
参考内容摘要	艾玛的音乐太棒了，真的非常喜欢这种充满魔力的风格。
零样本学习	我自己也是一名音乐家，已经从事这一行很多年了。而艾玛·布莱克里则是我最喜欢的艺术家。最喜爱的艺术家。
6万次微调	我自己也是一名音乐家，已经从事这个行业很多年了。虽然艾玛的音乐并非市面上那些最具技术性的死亡金属作品，我实在喜爱至极。她的音乐真的能打动人心。我。无论她演奏什么，无论是翻唱曲目还是原创作品，她的音乐总能触动我的心。在极其个人化的层面上。
监督学习 + 监督学习 + 6万次微调	我真的很羡慕艾玛·布莱克里的音乐，却不知道为何。 艾玛·布莱克里是我最喜爱的艺术家，她的音乐让我爱得无法自拔。我真希望自己能够随时聆听她的音乐。不过，她的音乐似乎在情感上给我带来了极大的伤害。 我觉得自己应该去欣赏她音乐中的人文内涵，而非技术层面的东西。我认为享受她的音乐应着眼于其中的人性因素，而非技术层面。她是我的最喜爱的艺术家。

表13：针对我们的TL;DR摘要生成任务生成的随机($T = 0.7$)个摘要，这些摘要的上下文是从TL;DR验证集中随机抽取的。

Samples for TL;DR summarization task (part 2)

context post	It really does. I did the first 5 seasons in a couple months, and ultimately I enjoyed them. The first three seasons cover just about all the really good stuff. Russell Edgington was, to me, the best supporting character in the show. He was everything that I valued about the show, which is to say, campy, violent, overly-gory fun. I went back to the show a little over a year later, and while S6 was tolerable with a few fun bits, S7 totally eroded. Tara’s plot goes full retard in the last season, people die for the sake of drama and convenience or leave with no real reverence for their history. Suddenly no one can make a good decision, and there was a needless romance bait-and-switch that ruined what I felt the show was moving towards. What’s terrible is that once you’ve done six seasons it’s hard to walk away, but each episode of seven stings more than the last. You don’t get that weird half-hollow/half-warm feeling you should when you finish a series of books or a long TV show. Outside of Pam and Erik, I didn’t give two fucks about any character by the end.
reference summary	Seasons one through three were a blast. Four was pretty meh, five and six weren’t too bad, seven sucked massively.
zero-shot	I finished S6 and still feel like I’m in the same place as I was with S5. Favorite Episodes: S1: "Tough Love" S2: "Crossover" S3: "The Party"
60k fine-tune	I did the first 5 seasons in a couple months, and ultimately I enjoyed them. The first three seasons cover just about all the really good stuff. Russell Edgington was, to me, the best supporting character in the show. He was everything that I valued about the show, which is to say, campy, violent, overly-gory fun
supervised	10/10 would watch again
supervised + 60k fine-tune	The first two seasons of the show were enjoyable. The show’s progress in the last three seasons is frustrating because it’s so similar to the first three seasons, and it’s so one-sided that there’s no reason to keep going. But the show’s greatest strength has always been its great writing, and it’s giving me that same feeling again.

Table 14: Random ($T = 0.7$) summaries for our TL;DR summarization task, with contexts randomly sampled from TL;DR validation set.

TL;DR摘要生成任务的样本（第二部分）

上下文帖子	确实如此。我仅用几个月时间就完成了前5季的内容，最终还觉得非常有趣。它们。前三季几乎涵盖了所有精彩的内容。拉塞尔·埃丁顿在我看来，埃丁顿是这部剧中最出色的配角。他具备了我所欣赏的该剧所有特质，也就是那种夸张、暴力且充满血腥场面的娱乐风格。也就是那种夸张、暴力且过度血腥的娱乐氛围。一年多后我再次观看了这部剧，虽然第六季还算勉强能看，有一些有趣的片段，但第七季已经彻底败坏了剧集质量。在最后一季中，塔拉的剧情线变得极其荒谬，角色们要么为了剧情需要而惨死，要么离开时对自身命运毫无敬意可言。为追求戏剧效果和便利而牺牲，最终离开时却对它们毫无真正的敬畏之心。剧情发展至此，突然间没人能做出正确的决定了，甚至还出现了一些不必要的浪漫情节。这种诱骗式套路破坏了原本我以为剧集正在朝着好的方向发展的态势。 糟糕的是，一旦演到了六季，就很难再放弃继续看了。不过每一集带来的痛苦都比上一集更强烈。你不会遇到那种奇怪的情况。也就是那种既空虚又略带温情的感觉。这是个电视节目。除了帕姆和埃里克之外，其他角色我根本没兴趣了解。end.
参考资料摘要	第一季到第三季都非常精彩。第四季就一般般，第五季和第六季也还算凑合。不过第七季糟糕透顶。
零样本学习	我已经看完了第六季，但感觉仍和看第五季时处于同样的水平。 最喜爱的剧集： 第一季：《严厉的爱》 第二季：《跨界联动》 第三季：《派对》
6万次微调	我在几个月内完成了前5个季数的任务，最终还挺享受这个过程的。第一个三个季数几乎涵盖了所有精彩的内容。对我而言，拉塞尔·埃丁顿就是这部剧中最出色的配角。他具备了我所看重的所有特质。展现出来，也就是说，那种夸张、暴力且血腥程度过高的娱乐内容。
监督学习 + 6万次微调	满分10分，绝对会再看一遍。 该节目的前两季相当精彩。而在最近的几季中，剧情的发展与前三季极为相似，这种单方面的重复让人毫无继续观看的兴趣。不过，这部剧最大的优势始终在于其出色的剧本，而如今它再次给我带来了同样的感受。一直以来都是它出色的文笔，此刻又让我感受到了同样的感受。

表14：针对我们的TL;DR摘要生成任务生成的随机($T = 0.7$)摘要，其中上下文是从TL;DR验证集中随机抽取的。

Samples for TL;DR summarization task (part 3)

context post	> you should be happy, lots of votes were counted twice in precincts that went to Clinton in a landslide, if they recount them by hand she will lose more votes, and probably a lot... You are assuming you understand what the problem here is and what the end result on the election was. The article does not state where the problem lies. E.g. The machine shows a tally of 120 votes having been processed on it. The actual reported vote tally to the election board was 100 votes. The 20 vote differential on the machine is due to 20 times a ballot needing to be re-scanned due to error. No double-counting of votes occurs but there is also a delta between what the machine says was recorded vs what was reported. In this case the final reported vote tallies may have been factually correct but there is no way to validate them against the machine record. To understand the problem an analysis and deconstruction of the machines and the election night events must be undertaken. Find out the cause of the problem, if the problem caused votes to be over/under counted, or if the problem was solely with machine data that was not used for tabulation of results.
reference summary	neither you nor I have any idea what the actual problem is so there is no basis for saying the problem favors either candidate at this time, or if either candidate would even be affected.
zero-shot	Election officials did not count ballots in a timely manner, by hand, to meet the threshold of getting votes counted on election night. Voters were not eligible to vote on election night and were not aware of the problems. The election officials did not realize that their system was broken until very late in the night.
60k fine-tune	The machine shows a tally of 120 votes having been processed on it. The actual reported vote tally to the election board was 100 votes. The 20 vote differential on the machine is due to 20 times a ballot needing to be re-scanned due to error. No double-counting of votes occurs but there is also a delta between what the machine says
supervised	Stop assuming the election was completely free of double-counting before you even begin to discuss the actual results. For any result to be close, it must be based on the total vote totals reported to the election board and not just the machine results. The election board did not report the machine results to the election board.
supervised + 60k fine-tune	The reported vote tallies may have been factually correct but there is no way to validate them against the machine record. The machine may have actually been accurate and the machine may have been faulty in other ways. The election may have been considered fair but the machine may not have been. The results are still in dispute.

Table 15: Random ($T = 0.7$) summaries for our TL;DR summarization task, with contexts randomly sampled from TL;DR validation set.

TL;DR摘要生成任务（第3部分）的样本

上下文帖子	> 您应该感到高兴，因为在那些最终归属克林顿的选区中，有许多选票被重复统计了。如果这些选票再通过手工方式重新统计，克林顿将会失去更多选票。 数量很可能非常多... ... 你以为自己已经明白这里的问题所在以及最终的选举结果会如何。不过该文章并未指出问题出在何处。 机器上显示已有120张选票被处理完毕。 E.g. 实际上，上报给选举委员会的选票总数仅为100张。机器上出现的20票差异，是由于有20张选票需要重新扫描所导致的。 20票机器上的差异是由于有20份选票需要重新扫描所致 出现错误。虽然不存在选票的重复计数问题，但系统记录的数值与实际报告的数值之间存在差异。即系统记录的数值与实际上报的数值不符。 在这种情况下，最终上报的投票总数可能在事实层面是正确的，但…… 由于无法以系统记录作为依据来验证这些数据，因此难以确认其准确性。 为了解决这一问题，必须对相关机器以及选举日当晚的相关事件进行深入分析和拆解。只有查明问题的根源，才能进一步采取相应措施。 该问题会导致投票数量被多计或少计，或者如果问题仅出在未被用于结果统计的机器数据上。那些未被用于统计结果的机器数据。
参考资料摘要。	无论是你还是我都不清楚问题的真正所在，因此根本无从判断。 目前还无法断言该问题会对哪位候选人有利，也不清楚任何一位候选人是否会因此受到影响。甚至还会受到影响。
零样本学习	选举官员未能及时手工统计选票，以符合在选举日当晚完成投票计数的要求。即要在选举日当晚就将所有选票计入的结果标准。 那些选民在选举日当晚没有投票资格，同时也并未意识到相关的问题。这些问题。 那些选举官员直到深夜时分才意识到他们的系统出现了故障。深夜时分。
6万次微调	机器显示已有120张选票被处理完毕，而实际上报给选举委员会的选票总数仅为100张。出现这20张选票的差异，是因为有20张选票因系统错误需要重新扫描。由于出现错误，该机器需要重新扫描20次投票数据。虽然会出现选票的重复计数现象，但机器计算出的结果与实际数值之间也存在差异。 says
监督学习	在开始讨论实际选举结果之前，先别急于认定此次选举完全没有发生重复计数的情况。只有基于选举委员会收到的全部投票总数，而非仅机器计算的结果，得出的数据才有可能较为接近真实值。这些数据指的是上报给选举委员会的总体投票数，而不仅仅是机器计算出的结果。选举委员会并未将机器计算出的结果提交给该机构。
采用监督学习方式 + 6万次微调	所公布的投票统计结果在事实层面或许是正确的，但却无法得到验证。无法将这些结果与机器记录进行对比核实。实际上机器的统计结果也可能是准确的。不过机器在其他方面可能存在故障，因此这次选举虽可能被视为公平的，但机器的统计结果未必可靠。尽管选举或许属于公平范畴，但机器的统计结果未必如此，相关结果至今仍存在争议。

表15：针对我们的TL;DR摘要生成任务生成的随机($T = 0.7$)个摘要，其中所使用的上下文是从TL;DR验证集中随机抽取的。

Fine-Tuning Language Models from Human Preferences		
Samples for CNN/DM summarization task (part 1)		
context article	Google launched a new U.S. wireless service today that switches between Wi-Fi and cellular networks Google is already the world’s most popular phone software provider, and a pay-TV operator - and now it wants to be your mobile network carrier. The company has unveiled a U.S. wireless service that switches between Wi-Fi and cellular networks to curb data use and keep phone bills low. The service, called 'Project Fi,' debuted today, about two months after Google revealed its plans to expand its ever-growing empire into providing wireless connections for smartphones. Google is selling the basic phone service for \$20 a month and will only charge customers for the amount of cellular data that they use each month, instead of a flat rate. Each gigabyte of data will cost \$10 a month. That means a customer could sign up for a plan offering three gigabytes of data and get \$20 back if only one gigabyte was used in a month. [...<i>Truncated</i>...] Rather than building its own network, Google is leasing space on cellular towers built by Sprint and T-Mobile, which are hoping the deals will boost their profits without costing them too many customers tempted to defect to Project Fi.	
reference summary	Project Fi will be hosted through Sprint Corp and T-Mobile’s networks. It costs \$20 for basic service and unused data is paid back to customer. The invitation-only service will work only on Nexus 6 phones in the US. Numbers will live in the cloud so users can talk on any connected tablet.	
zero-shot	Project Fi is a new smartphone service that switches between Wi-Fi and cellular networks to curb data use and keep phone bills low. The service is available for an introductory price of \$20 a month and will only charge customers for the amount of cellular data that they use each month.	
60k fine-tune	Google is already the world’s most popular phone software provider, and a pay-TV operator - and now it wants to be your mobile network carrier. The company has unveiled a U.S. wireless service that switches between Wi-Fi and cellular networks to curb data use and keep phone bills low.	
supervised	Google launched a new U.S. wireless service today. Called 'Project Fi,' it switches between Wi-Fi and cellular networks. The service, called 'Project Fi,' debuted today. About two months after Google announced its plans to expand its ever-growing empire into providing wireless connections for smartphones.	
supervised + 60k fine-tune	Google launched a new U.S. wireless service today that switches between Wi-Fi and cellular networks. The service, called 'Project Fi,' debuted today, about two months after Google revealed its plans to expand its ever-growing empire into providing wireless connections for smartphones.	

Table 16: Random ($T = 0.5$) summaries for our CNN/DM summarization task, with contexts (shown truncated) randomly sampled from CNN/DM test set.

基于人类偏好对语言模型进行微调		
CNN/Daily Mail数据集摘要生成任务样本（第一部分）		
上下文艺术- cle	谷歌今日推出了全新的美国无线服务，该服务可在Wi-Fi与蜂窝网络之间切换。蜂窝网络 谷歌已然成为全球最受欢迎的手机操作系统提供商，同时它也在... 付费电视运营商——而现在它还想成为你的移动网络服务商。 该公司推出了一项美国无线服务，可在Wi-之间切换。 利用Fi及蜂窝网络来控制数据使用量，从而降低电话账单。 名为“Project Fi” 概这项服务在谷歌推出约两个月后的今日正式上线。 公布了其将不断扩张的业务版图延伸至无线领域的计划 智能手机的连接接口。 谷歌目前以每月20美元的价格提供基础电话服务，且仅会按照用户每月使用的蜂窝数据量收费，而非收取固定费用。 也就是说，用户无需为固定额度付费，只需根据实际使用的数据量来结算费用。 每GB的数据每月的费用为10美元。这意味着用户有可能…… 选择套餐获得3GB的数据用量，即便一个月内仅使用了1GB，也能获得20美元的退款。 在一个月内就被用完了。 [...已截断的内容： ...] 相比自行建设网络，谷歌正在租赁蜂窝网络上的频谱资源。 这两家公司期望通过这些合作提升利润，同时避免因太多客户转而使用Project Fi服务而遭受损失。 同时避免因流失过多客户而影响那些想转用Project Fi服务的用户。 Project Fi服务将通过Sprint Corp公司和T-Mobile公司的网络来提供，其订阅费用为20美元。 基础服务会收费，未使用的数据则会退还给客户。此服务仅限受邀用户使用。 仅在美国的Nexus 6手机上可用。相关数据将存储在云端，以便用户使用。 可以在任何联网的平板电脑上进行对话。	
参考资料 摘要		
零样本学习	Project Fi服务是一种全新的智能手机服务，它能够在Wi-Fi网络和蜂窝网络之间自动切换，从而控制数据使用量并降低手机话费。 该服务目前的起步价为每月20美元，用户只需为其每月实际使用的蜂窝数据量付费。 此外，这项服务还会进行6万次微调，以进一步提升性能。 谷歌早已是全球最受欢迎的手机操作系统提供商，同时还是一家付费电视服务提供商。	
P值	谷歌已然成为全球最受欢迎的手机操作系统提供商，同时也是一家付费电视服务商。 运营商——如今它还想要成为你的移动网络服务提供商。 这家公司推出了一项美国无线服务，该服务能够在Wi- Fi与蜂窝网络之间切换。 通过这种方式，它旨在控制数据使用量，从而降低手机话费。	
监督学习	谷歌今日推出了一项新的美国无线服务。这项名为“Project Fi”的服务能够在不同网络间切换。 也就是说，它可以在Wi-Fi和蜂窝网络之间进行切换，而这项名为“Project Fi”的服务今天正式上线了。 在谷歌宣布计划进一步扩展其不断扩张的业务版图大约两个月后 该计划旨在为智能手机提供无线连接服务。	
监督学习 + 6万次微调	今日，谷歌推出了一项新的美国无线服务，该服务能够在Wi-Fi与蜂窝网络之间切换。这项名为“Project Fi”的服务就在此前大约两个月后于今日正式亮相。 蜂窝网络。这项名为“Project Fi”的服务在大约两个月后于今日正式推出。 谷歌公布了其计划，要将不断扩张的业务版图进一步延伸至为智能手机提供无线连接服务。 智能手机的连接接口。	

表16：针对我们的CNN/Daily Mail摘要生成任务生成的随机（ $T = 0.5$ ）摘要，其中所使用的上下文（仅显示部分内容）是从CNN/Daily Mail测试集中随机抽取的。

Fine-Tuning Language Models from Human Preferences	
Samples for CNN/DM summarization task (part 2)	
context article	With the date of their hugely anticipated bout less than a month away, Floyd Mayweather and Manny Pacquiao’s training intensity has hit gruelling levels ahead of the clash in Las Vegas. Mayweather is as renowned for showing off his extravagant lifestyle as he is for going undefeated in his professional career. But the latest pictures released from his training camp, appearing to show him hanging a weight from his head, displays the extreme level he hits in order to be as fit as possible for his fights. Manny Pacquiao (right) poses alongside actor Mark Wahlberg, who visited the Filipino at training Floyd Mayweather released two pictures from his training camp appearing hold a weight from his head Mayweather (left) takes on Pacquiao in a £160million mega-fight in Las Vegas on May 2 Mayweather (left) appeared to take his training intensity to a new height before his meeting with Pacquiao As for Pacquiao, he had a famous visitor check up on his preparations for May 2. [...Truncated...] Pacquiao and Mayweather’s clash will be a reported £160million mega-fight, with Pacquiao seeking to inflict a first ever defeat upon Mayweather. The 36-year-old Pacquiao also uploaded a picture to his Twitter page away from training with his daughter.
reference summary	Manny Pacquiao took time out from training to meet Mark Wahlberg. Floyd Mayweather shared a picture of him holding a weight with his head. The due meet in Las Vegas in a £160million encounter on May 2.
zero-shot	Mayweather and Pacquiao are going to clash in Las Vegas on May 2.
60k fine-tune	With the date of their hugely anticipated bout less than a month away, Floyd Mayweather and Manny Pacquiao’s training intensity has hit gruelling levels ahead of the clash in Las Vegas. Mayweather is as renowned for showing off his extravagant lifestyle as he is for going undefeated in his professional career.
supervised	Floyd Mayweather and Manny Pacquiao will fight in Las Vegas on May 2. Pacquiao has a visitor check up on his preparations for the bout. Mayweather released pictures of his training camp to social media.
supervised + 60k fine-tune	Floyd Mayweather and Manny Pacquiao will fight in Las Vegas on May 2. Mayweather released two pictures from his training camp appearing hold a weight from his head. The 36-year-old boxer is as renowned for showing off his extravagant lifestyle as he is for going undefeated in his professional career.

Table 17: Random ($T = 0.5$) summaries for our CNN/DM summarization task, with contexts (shown truncated) randomly sampled from CNN/DM test set.

基于人类偏好对语言模型进行微调	
CNN/Daily Mail数据集摘要生成任务样本（第二部分）	
上下文艺术- cle	在他们备受瞩目的对决日期距离现在不到一个月之际，弗洛伊德·梅威瑟与曼尼·帕奎奥的训练强度已提升到极为艰苦的程度。拉斯维加斯的那场冲突。 弗洛伊德·梅威瑟同样以炫耀其奢华的生活方式而闻名。为在其职业生涯中保持不败纪录。 不过，最新从他的训练营流出的照片显示…… 那些照片似乎展现出了他训练时的情景。照片中他正把重物挂在头上，这充分体现了他为保持最佳状态、以便在比赛中取得优势而付出的极端努力。所有这些努力都是为了在比赛时处于最健康的身体状态。 曼尼·帕奎奥（右）与演员马克·沃尔伯格一同摆出姿势。谁访问过菲律宾语训练 弗洛伊德·梅威瑟公开了两张来自其训练营的照片，照片中他正将重物举过头顶。这些照片展示了他正在进行相关训练的画面。 梅威瑟（左）将与帕奎奥在拉斯维加斯进行一场价值1.6亿英镑的超级对决。5月2日的比赛加油！ 弗洛伊德·梅威瑟（左）在与曼尼·帕奎奥的对战之前，似乎将训练强度提升到了新的高度。在与曼尼·帕奎奥会面之前 至于曼尼·帕奎奥，有一位著名的访客前来视察他的备战情况。5月2日。 [..已截断的内容..] 据报道，曼尼·帕奎奥与弗洛伊德·梅威瑟的这场对决将是一场价值1.6亿英镑的超级大战。帕奎奥的目标是在这场比赛中为梅威瑟带来史上首次败绩。 36岁的帕奎奥还在自己的推特主页上上传了一张照片，照片内容是他正在和女儿一起训练。
参考资料摘要	曼尼·帕奎奥暂时暂停训练，前去与马克·沃尔伯格会面。弗洛伊德·梅威瑟——还分享了一张他举着重物、低着头的身影照片。两人定于5月2日在拉斯维加斯进行一场价值1.6亿英镑的较量。
零样本学习 6万次微调	梅威瑟与帕奎奥将于5月2日在拉斯维加斯展开对决。 距离他们备受瞩目的对决日期不到一个月，弗洛伊德·梅威瑟与曼尼·帕奎奥在本次拉斯维加斯之战前的训练强度已达到极为艰苦的程度。拉斯维加斯的那场冲突。 弗洛伊德·梅威瑟不仅以职业拳击生涯中保持不败的战绩闻名，更以其奢华的生活方式而广为人知。
监督学习	弗洛伊德·梅威瑟与曼尼·帕奎奥将于5月2日在拉斯维加斯展开对决。帕奎奥有访客前来查看他为这场对决所做的准备情况。弗洛伊德·梅威瑟发布了相关照片将他训练营的点滴上传到社交媒体上。
监督学习 + 6万次微调	弗洛伊德·梅威瑟与曼尼·帕奎奥将于5月2日在拉斯维加斯展开对决。梅威瑟发布了两张来自他训练营的照片，照片中他似乎正用重物举过头部。这位36岁的拳击手不仅因在职业比赛中保持不败纪录而闻名，还因其奢华的生活方式备受瞩目。为在其职业生涯中保持不败纪录。

表17：针对我们的CNN/DM摘要生成任务生成的随机($T = 0.5$)摘要，其中上下文（仅显示部分内容）是从CNN/Daily Mail测试集中随机抽取的。

Samples for CNN/DM summarization task (part 3)

context article	The Solar Impulse 2, the experimental plane attempting to fly around the world without using a drop of fuel, has been grounded by the weather in China. What was supposed to be an overnight pit stop in the southwestern city of Chongqing has now stretched into a two-and-a-half week stay. It’s an unexpected layover that is testing the patience of the pilots and more than 60 team members, just a quarter of the way along a journey that will cover some 35,000 kilometers (21,748 miles) over five months. "You can believe that your will can influence the weather," says pilot Bertrand Piccard, who comes from a long family line of explorers. But "then you’ll be really disappointed because it doesn’t work. "Or you just know that it’s an adventure you do with the forces of nature." Piccard, along with Andre Borschberg, a Swiss engineer, are taking turns flying the single-seater aircraft, which took off from Abu Dhabi on March 9 and has successfully flown through Oman, India, and Myanmar. There is much at stake. [...<i>Truncated</i>...] "Our boss is the sun," says Solar Impulse spokeswoman Claudia Durnat. Durnat says there may be a tiny window ahead, with the next possible departure from Chongqing on Tuesday. "It’s not good before, and the days after don’t look very clear."
reference summary	Solar plane attempting to be first to circumnavigate world without using fuel is stuck in China. Solar Impulse 2 attempts to prove the power of renewable energy.
zero-shot	The plane has been grounded in China for two weeks because of bad weather.
60k fine-tune	The Solar Impulse 2, the experimental plane attempting to fly around the world without using a drop of fuel, has been grounded by the weather in China. What was supposed to be an overnight pit stop in the southwestern city of Chongqing has now stretched into a two-and-a-half week stay.
supervised	Solar Impulse 2 has been grounded by the weather in China. The plane took off from Abu Dhabi on March 9. The plane is trying to prove the power of renewable energy.
supervised + 60k fine-tune	Solar Impulse 2 has been grounded in China for two-and-a-half weeks. The plane is attempting to fly around the world without using a drop of fuel. The team, which includes Bertrand Piccard, is taking turns flying the single-seater. The plane took off from Abu Dhabi on March 9 and has successfully flown through Oman.

Table 18: Random ($T = 0.5$) summaries for our CNN/DM summarization task, with contexts (shown truncated) randomly sampled from CNN/DM test set.

CNN/Daily Mail数据集摘要生成任务示例（第3部分）

上下文艺术- cle	试图在不使用一滴燃料的情况下环球飞行的实验性飞机——太阳脉冲2号已被中国恶劣天气迫使停飞。 原定在西南部城市进行的过夜中途停留 在重庆的行程现已延长至两周半。 这是个意外的中途停留，正在考验着飞行员们的耐心，甚至更多。 超过60名团队成员，而这仅仅是在一段漫长旅程中已完成四分之一的路程，这段旅程还将涵盖更多内容 五个月内飞行35,000公里（21,748英里）。 “你可以相信，人的意志能够影响天气，”飞行员贝特朗·皮卡尔说道。他出身于一个世代为探险家的家族。但“那样的话你肯定会非常失望的”因为它根本无法正常运行。 “或者说，你只是知道这是一场与大自然力量共同进行的冒险。” 皮卡尔与瑞士工程师安德烈·博尔施伯格轮流驾驶这架单座飞机——该飞机于3月9日从阿布扎比起飞，至今已 成功飞越了阿曼、印度和缅甸。 成败关乎重大。 [……已截断的内容……] 阳光动力号的发言人克劳迪娅·杜尔尼亚特表示：‘我们的领航者就是太阳。’ Durnat表示，前方或许还有一段短暂的窗口期，即下一次可能的脱离时刻。 ‘情况非常不明朗。’ 非常清晰。
参考资料 摘要	那架试图成为首架无需燃料即可环游世界的太阳能飞机目前陷入了困境。在中国，太阳脉冲2号正努力证明可再生能源的巨大潜力。
零样本学习 6万次微调	由于恶劣天气，这架飞机已在中国停飞两周时间。 试图在不使用一滴燃料的情况下飞遍全球的实验性飞机——阳光动力2号，因中国境内的恶劣天气而被迫停飞。 原本计划在中国西南部城市重庆进行的一次过夜检修，如今已延长为两周半的停留时间。 ‘监督学习’是指在人工干预下对模型进行训练的学习方式。 阳光动力2号正是由于中国境内的天气状况而不得不暂停飞行任务的。这架飞机是从……起飞的。
监督学习	由于中国境内的天气状况，阳光脉冲2号不得不停飞。这架飞机是从……起飞的3月9日，飞机正在阿布扎比尝试证明可再生能源的巨大潜力。
监督学习 + 6万次微调	太阳脉冲2号已在中国停飞两个半星期，这架飞机正试图在不使用任何燃料的情况下环游世界。这支队伍正努力完成这一壮举，其中就包括贝特朗·皮卡尔，他们轮流驾驶这架单座飞机。该飞机已经成功起飞。 该数据来自3月9日的阿布扎比，且已顺利飞越阿曼。

表18：针对我们的CNN/DM摘要生成任务生成的随机（ $T = 0.5$ ）摘要，其中所使用的上下文（仅显示部分内容）是从CNN/Daily Mail测试集中随机选取的。